Consistency theory of SMC-BVS (Wave 1 closure)

Posterior, sampler, reported-model and delivery results for $p \leq n^A$, with the Song–Liang (2015) map

For Felipe Bedoya and co-authors

Compiled by the Wave 1C writer (Claude Opus 5.5)
from the Wave 1 / Wave 1C record

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For agents — how to read this chapter

- This is a *statement* document: every theorem is transcribed from the proof package named in its status box; proofs are **not** reproduced. Hypotheses are written out so that the statements can go into the paper; the proof files and the review trail are given for each result (§9).
- Status words are copied verbatim from the cited files and are never upgraded. "PROVED-CHECKED" is a Wave-1/1C ticket status: proof file + cross-family lemma checks + independent mid-point CONTINUE + exhaustive sweep fixed + two-seat council passed (both ≥ 8.0, all gates, same round) [W1C/PROTOCOL.md:35]. These ticket results are *not yet adopted by HQ into the frozen registry*; registry statuses of the State-of-Project (SoP) cards are quoted separately where they enter.
- The only new mathematics in this chapter is Lemma 7.1 (§7.1, the LS0-cap integration lemma). Its status is PROVED (by this seat); checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6), with one cross-family check (fidelity reviewer F1, Codex gpt-6.1-sol high: VERIFIED-WITH-AMENDMENTS, amendments applied, `W1C/FINAL/reviews/F1_fidelity_sol.md`); the round-1 and round-2 council seats checked it and found it correct, neither round passed; the final council (round 3: Sol 9.3, Opus 8.6, all six gates PASS) passed the chapter including it. Proposition F in A1's corrected form and its GROW extension (A2, checked by A3) are HQ-audit results (A1 HOLDS-WITH-AMENDMENTS; PROVED (A2), checked by A3); checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6). Round-1 council findings and their disposition: `W1C/FINAL/reviews/C1_disposition.md`.
- Path aliases in citations: `W1/` = `Theory_work/Wave1/`, `W1C/` = `Theory_work/Wave1C/`, `SoP/` = `Theory_work/State_of_Project/sections/`, `CORE/` = the frozen study core `Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/`, `PROD/` = `Scripts/SMC_BVS/core/`, `BR/` = a file as committed on branch `origin/w1c/ls0-merge` (C-ENG, reviewed head `8a27652`; checked with read-only `git show`), `BRH/` = the same file at the current head `493579a`. Every `path:line` citation was checked by `W1C/FINAL/citecheck.py` (output `W1C/FINAL/CITECHECK.txt`).

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1 Setting, method and notation

Full definitions are in `SoP/02_method.tex` (method v3.1) and `SoP/A_notation.tex`; this section fixes only what the theorems below use.

1.1 Data, target posterior, start

Model G. Fixed (or, where stated, random) raw design $X^r \in \mathbb{R}^{n\times p}$ with nonconstant columns; $y^r = \alpha\mathbf{1} + X^r\beta + \varepsilon$, $\varepsilon \sim N_n(0, \sigma^2 I_n)$, $\sigma^2 > 0$ fixed; in random-design statements ε is independent of X^r. Columns are centred and scaled, $X = CX^r D_x^{-1}$ with $C = I - \mathbf{1}\mathbf{1}^\top/n$ and $\|x_j\|^2 = n-1$; the response is centred and scaled inside the score. $\mu := CX^r\beta = X\theta^*$, $\theta^* := D_x\beta$, $S_0 := \operatorname{supp}\beta$, $s_0 := |S_0| \geq 1$ fixed, $p = p_n \leq n^A$ for a fixed $A \geq 1$ ($p \gg n$ allowed) [SoP/04_theory.tex:70-74]; [W1C/C-GROW/STATEMENT.md:56-63].

Target (D0). $K = \min(p, \lfloor n/2 \rfloor)$, $\Omega = \{S : |S| \leq K\}$. With $G_U = X_U^\top X_U$, $z_U = X_U^\top y$, $H_U = (n+1)G_U + I$, $Q_U = y^\top y - n z_U^\top H_U^{-1} z_U$, $\nu = 0.01 + n/2$:

$$m(U) = c_n\Big[\frac{\det H_U}{\det(G_U+I)}\Big]^{-1/2}(0.02 + Q_U)^{-\nu}, \qquad w(k) = \Big[2\binom{2p-1}{k}\Big]^{-1}, \qquad \pi(S) \propto w(|S|)\, m(S) \;\; (S \in \Omega),$$

the shipped $g = n$ ridge score with IG(0.01, 0.01) and the capped BB$(1, p)$ size prior [SoP/04_theory.tex:71-72]. $F(S) := \log w(|S|) + \log m(S)$. $\mathrm{PIP}_j := \pi(j \in S)$, $\mathrm{MPM} := \{j : \mathrm{PIP}_j > \frac{1}{2}\}$, $\xi_q := \{j : \mathrm{PIP}_j > q\}$ for $q \in (0,1)$, and the *exact-posterior* $\mathrm{HPM} := \arg\max_\Omega \pi$ (distinct from the method's reported model $\hat{S}^+$ below) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:11-18].

Blocks, start law, start mode. The method partitions the columns into blocks $C_1, \ldots, C_J$ (complete linkage on $1 - |\hat{r}|$, top-down cut at $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$, next-fit packing; a deterministic function of X). Block tables $r_b(U) = m(U)/m(\emptyset)$; start $\gamma_0(S) \propto w(|S|)\prod_b r_b(S \cap C_b)$ on Ω, $\pi_0 = \gamma_0/\bar{Z}_0$; the **start mode** $\tilde{S}_0 = \arg\max_\Omega \gamma_0$ is computed exactly by a block DP (`CORE/cpp/v3_start.h`) [W1/W1-GEN/STATEMENT.md:26-38].

1.2 Method v3.1 + LS0, reported model, output event

Local search (M §6.10). From a centre, scan every feasible flip (additions beyond K excluded) and every swap with the centre frozen; every scanned model is scored; move to the best scanned neighbour if its F is *strictly* larger than the centre's (exact F ties among neighbours broken lexicographically); at most ℓ moves, then one further scan; `local_limit` iff that scan still improves; `local_certified` iff the returned model is a scanned centre with no improving neighbour [SoP/02_method.tex:463-467]; code [CORE/cpp/v3_engine.h:376-425]. Production cap $\ell_n = \lceil \log_2(n + p + 1) \rceil$.

LS0. After the final local search (and after the extra search from c_4 when $R' > 4$), one local search with the §6.10 rules from the start mode $\tilde{S}_0$; LS0 reads no uniforms, every model it scores joins the all-scored set and the stores, its centre is scored (emitted); a completed LS0 capped at `local_limit` still returns output [SoP/02_method.tex:490-501]. LS0 is **not** in the frozen production core (which calls only the search from the global best, [PROD/cpp/v3_core.cpp:447]); C-ENG implemented it on branch `w1c/ls0-merge` (default on, switch `ls0`; not merged to main; §7).

Reported HPM. $\hat{S}^+$ is the maximum of the all-scored set $\mathcal{S}^+$ (every model scored by the fit up to and including LS0, post-hoc certificate evaluations excluded) for the strict total order $(F(S), -|S|, \text{lex})$ [SoP/02_method.tex:468-473].

Output event and potential run. $\mathcal{O}$ is the event that the fit returns estimates (status `exact` or `completed`); $U_n := \mathcal{O}^c$. Guarantees are stated for the potential uninterrupted run and as $\mathbb{P}(\mathcal{O} \cap \cdot)$, never as $\mathbb{P}(\cdot \mid \mathcal{O})$ [SoP/02_method.tex:579-583]. Bars denote potential-run objects; the never-scored term is

$$D_n := \mathbb{P}\big(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\}\big) \quad \text{(potential-run notation; the bar is not a complement)}$$

[SoP/04_theory.tex:84].

1.3 Named implementation premises

The reported-model theorems are proved for the method *as specified* under named premises; these are not model assumptions and are not discharged by the theorems [SoP/04_theory.tex:84]: **EX-A** exact scores and arithmetic (an idealisation, not a validated floating-point property); **EX-S** fresh independent ideal uniforms; **CAL** a successful frozen calibration record on streams 0–9; **IND-T** final streams independent, $R' = 4k$ declared before stream 10 (needed when $R' > 4$); **GUARD-W** stop-only guard (operational interruption ⇒ no output); **SR** the stop/return rule; **OP-LS0** LS0 integrated as specified; **e4** the emitted LS0 centre is the prescribed smallest-size, lexicographically smallest start-mode maximiser (DP invariant N3 OPEN) (for PXN, GEN and M5(b1-LH) replaced by the weaker **e4′**, §5.2; for GROW by 'the emitted centre is some maximiser of γ_0', which the coded routine satisfies by W1-N3(a); delivery keeps e4 with TDP-IMPL); **complete scans**. The delivery results (§6) add the idealisation premises **IDEAL-W** (unbounded words), **MACH** (wall time and bytes proportional to operations and words; allocations succeed within the declared memory; no external interruption), **CONFIG** (production core plus LS0 with cap ℓ_n and the e4 fallback, default declared list, polynomial certificate budgets, fixed worker count) and **TDP-IMPL** (the e4 tie fallback is W1-N3's reference algorithm TDP), and do *not* assume CAL. Two different method requirements carry the label MR1: **MR1 (C-PACK)** (LS0 centre = coded start mode) and **MR1 (C-GROW)** (LS0 move cap $\ell_n + |\tilde{S}_0|$) [W1C/C-DLV/STATEMENT.md:21-46].

1.4 Three layers

As in the SoP regime map [SoP/04_theory.tex:822], results live on three layers that compose only explicitly: **(i)** the exact posterior π concentrates on S_0 (§3); **(ii)** the sampler's outputs track π or $h(S_0)$ (§4); **(iii)** the reported model $\hat{S}^+$ equals S_0 jointly with output (§5). Delivery, $\mathbb{P}(\mathcal{O}) \to 1$, is a fourth question (§6). M5(0) composes (i) and (iii) at every n:

$$\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2\,\mathbb{E}_0[1 - \pi(S_0)] + D_n$$

(the deterministic part: if $S_0 \in \mathcal{S}^+$ and $1 - \pi(S_0) < \frac{1}{2}$ then $\hat{S}^+ = S_0$), status T38-(0) "PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))" [SoP/04_theory.tex:477-489]. Hence every reported-HPM theorem below reduces to $\mathbb{E}_0[1 - \pi(S_0)] \to 0$ (layer (i)) plus $D_n \to 0$ (discovery of S_0 by the search).

2 The Song–Liang (2015) map

Song & Liang (2015, JRSS-B 77(5), 947–972) is the template for the story: posterior consistency, sure screening, thresholded and MAP-model consistency, and a misspecification theorem. The table updates C-PACK's map v2 [W1C/C-PACK/SONG_LIANG_MAP.md:1-12] with C-MIS, C-GROW and C-DLV/A1. Two global differences: (a) their correctly specified asymptotics also assume $P_n/n^\theta \to \infty$ for some $\theta > 0$, while ours impose $p_n \leq n^A$ (LH-iid: also $p_n \to \infty$) with no positive-power lower growth bound — different domains, not identified [W1C/C-PACK/SONG_LIANG_MAP.md:10-12]; (b) their Theorem S is in an online supplement that was **NOT INSPECTED** (publisher access failed); comparison is with the main-text description only [W1C/C-PACK/REPORT.md:87] .

Song–Liang	their content	our analog (this chapter)	status (verbatim)
Thm 1 (p. 952), posterior consistency	Hellinger contraction, exponential inner/outer rates, $\|x_i\| \le 1$ (a uniform bound on each predictor), approximate sparsity	**No Hellinger/predictive analog.** Selection analog: Thm 3.1 (S-T/T03♯), $1-\pi(S_0) = O_P(\sqrt{\log n/n})$, fixed design, $p \le n^A$	T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2); Hellinger: CJ20 FW20 Not delivered; selection counterpart in S-T only
Thm 2 (p. 953)	random design, Condition 1, strict sparsity; Hellinger	random-design *selection* analogs: PXN (P) on LH-iid (joint law); GEN class via (C4), instance I-c (Cor. 3.3)	W1-PXN, W1-GEN PROVED-CHECKED (Wave 1, not adopted by HQ)
Cor 1 (p. 954)	$P_n < \exp(Cn^\alpha)$, two prior branches, rates	rate $\sqrt{\log n/n}$ for $p \le n^A$ only; exponential p not covered	CJ18 No completed prior-tail/contraction route; p ≤ n^A stays in scope
Cond 2, (16) (p. 954)	second moments ⇒ convergence in probability with exponential inner threshold	all our statements are O_P with a polynomial rate; no exponential or expectation rate	— (comparison of modes)
Thm 3, sure screening (p. 956)	per true predictor, $\mathbb{P}^*[\pi(t \in \xi) < 1-e^{-c_1 n\varepsilon_n^2}] < e^{-c_1 n\varepsilon_n^2}$ (sensitivity only)	SL0(i), SL1(a): $\max_j \|\mathrm{PIP}_j - 1\{j \in S_0\}\| \le 1-\pi(S_0) = O_P(\sqrt{\log n/n})$ — sensitivity *and* specificity, uniformly in j; SL1′ without beta-min (large signals, sensitivity only)	PROVED (by this seat, unchecked); R34 re-derived; C-PACK council passed
ξ_q (p. 956)	"$\pi(t \subset \xi_q) \to_p 1$" if all true predictors are identifiable (read: $\mathbb{P}^*(t \subseteq \xi_q) \to 1$)	SL1(b): $\mathbb{P}(\xi_{q_n} = S_0) \to 1$ (equality) for fixed $q \in (0,1)$ or deterministic $q_n \in (0,1)$ with $r_n = o(\min(q_n, 1-q_n))$; exact MPM consistent	as row above
Thm S, MAP (supplement)	"establishes the consistency of MAP models" — exact hypotheses **NOT INSPECTED**	exact HPM: SL0(iv), SL1(c). **Reported $\hat{S}^+$**, jointly with output: LH-iid (Thm 5.2), GEN class (Thm 5.3), growing relevance sets incl. study S2 (Thm 5.4), misspecified candidate sets (Cor. 5.6); without the output qualifier (ideal machine, amended allocation) $\mathbb{P}(\hat{S}^+ = S_0) \to 1$ for the patched calibration on LH, LH-iid, GEN class (Prop. 6.3) and the GROW class (Prop. 6.4)	W1-PXN, W1-GEN, C-GROW, C-MIS PROVED-CHECKED; C-DLV PARTIAL; Prop. F with cap ℓ_n (C-DLV form): PROVED-CHECKED for the patched calibration law; corrected form with cap a_n (Prop. 6.3): A1 HOLDS-WITH-AMENDMENTS; checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6); GROW extension (Prop. 6.4): PROVED (A2), checked by A3; checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6)
SAM inclusion probabilities (§3.2, §7)	importance-weighted estimates; not itself Thm 3	islandwise weighted bound (Cor. 4.2); stacking route M1(i); store-restricted SL2. **No in-probability sure-screening theorem for the shipped pooled $R' = 4$ PIPs**	Cor. E PROVED under the same premises; M1(i) (h), (i): PROVED (by this seat, unchecked)
Cond 3, identifiability (p. 955)	uniform exponential-moment identifiability	$\mathrm{SE}_\lambda/\mathrm{SE}_w$ + beta-min; GEN's Gram floors; GROW's (G2)–(G3) — related role, *not* asserted equivalent	— (comparison)

Song–Liang	their content	our analog (this chapter)	status (verbatim)
Thm 4, misspecified subset (p. 957)	fixed candidate subset **s**, Condition 4 (finite second moments of $a^\top x_{\mathbf{s}}$, $\lvert a_i\rvert \le \delta$), prior positive on all models; conclusion (18): a.s. posterior consistency at the KL projection f_0	Thm 3.5: *exact-model* concentration $\pi_{\mathcal{C}}(S_0^{\mathcal{C}}) \to 1$ at rate $\sqrt{\log n/n}$, $p_{\mathcal{C}} \le n^A$, under the Gaussian-residual condition (GR) (stronger hypotheses); Thm F (fixed design, exact projection). No density analog	C-MIS PROVED-CHECKED
Cor 2, (19) (p. 957)	$\pi(\tilde{s} \subseteq \xi \mid D_n^s) \to_P 1$, $\tilde{s} = \operatorname{supp}\beta_0$ (screening); withdrawn under a binding cap	MIS no-beta-min clause; Lemma GD(a) recovers Cor. 2's conclusion in the jointly Gaussian fixed-$p_{\mathcal{C}}$ case, with a rate; Prop. P neighbour rule	C-MIS PROVED-CHECKED
Cond 5, $\lvert\operatorname{cov}(x_t, y)\rvert > 0$ (p. 958)	keeps every true predictor in SAM's screened set	analogy only: GEN's certificate (C1′) and GROW's block visibility (TV) are block-level conditions; masked truths (zero block coefficient) are recovered by the descent (GEN)	GEN PROVED-CHECKED; (C1′) VERIFIED-WITH-AMENDMENTS (R32)
§6, time (p. 968)	parallel time with $K_n \asymp \sqrt{P_n}$ processors	M8 accounting remark only; no speed claim. Delivery under a declared polynomial allocation: Prop. 6.3	M8 assembly PROOF SKETCH (by this seat, unchecked)

Sources: [W1C/C-PACK/SONG_LIANG_MAP.md:14-28]; [W1C/C-MIS/STATEMENT.md:15-35]; [W1C/C-MIS/REPORT.md:3-9]; [W1C/C-GROW/REPORT.md:24-30]; [W1C/HQ_AUDIT/A1_propF_redteam.md:3].

Plain reading — the map in one paragraph

Every correctly specified Song–Liang *selection* result (Thm 3, ξ_q, MAP) has an exact-posterior analog that is a one-line consequence of $1-\pi(S_0) \to 0$ (Lemma SL0, deterministic), with rate $\sqrt{\log n/n}$ for $p \le n^A$ and with specificity added. Beyond Song–Liang, the method's *reported* model is proved consistent jointly with output on a general correlated class (GEN), on growing relevance sets including the study's AR(1) design (GROW), and under misspecification (C-MIS Cor. D), and, on LH, LH-iid, the GEN class and the GROW class, with a total tier-1 calibration fallback, also without the output qualifier, on an ideal unit-cost machine with a declared polynomial allocation (Props. F and F-GROW: IDEAL-W, MACH, amended allocation; not checked for the C-MIS setting). Gaps: Hellinger/predictive contraction and exponential rates; exponential p; shipped pooled $R' = 4$ PIPs; Theorem S's hypotheses not inspected; density-level misspecification.

3 Posterior level

3.1 Posterior consistency with rate (S-T, T03♯)

Theorem 3.1 (S-T / T03♯; card TH09). *Fixed-design triangular array (model G), fixed $s_0 \ge 1$, $p \le n^A$ for a fixed A, $s_0 \le K$ eventually. Assume*

- ***R**: $\mu = X\theta^*$, $\|\mu\|^2 \le Mn$, and $\lambda_{\min}(\hat{R}_{S_0}) \ge \phi_0^2 > 0$ ($\hat{R} = X^\top X/(n-1)$);*
- ***tail cutoff**: $R_m = \sum_{k=\max(m+1,0)}^{K} \binom{p}{k} w(k)/w(s_0)$, $m_n^\sharp = \min\{m \ge s_0 : R_m \le n^{-(s_0+2)/2}\}$ (deterministic, data-free);*
- ***SE**$_\lambda(j_0)$, $j_0 = \lceil (2A+1)s_0 \rceil + 1$: every $S_0 \cup J$, $\lvert J\rvert \le j_0$, has normalised minimum eigenvalue $\ge \phi_0^2$; **SE**$_w(m_n^\sharp; g_0)$: every $S_0 \cup J$, $\lvert J\rvert \le m_n^\sharp$, has unnormalised minimum eigenvalue $\ge g_0 > 0$;*
- ***beta-min**: BM-∞, $\theta_{\min}^{*2} n\phi_0^2/(\sigma^2 \log n) \to \infty$; or BM($c$) with a fixed $c > c_0(s_0, A)$, meaning $\theta_{\min}^{*2} \ge c\sigma^2 \log n/(n\phi_0^2)$ eventually, $c_0 = \max\{4C_*, 32[s_0(A+1)+2]\}$ (C_* the locked T03 proof constant).*

Then $1-\pi(S_0) = O_P(\sqrt{\log n/n})$*; the exact HPM and MPM are consistent. Without beta-min, for* $J_n(c) = \{j \in S_0 : \theta_j^{*2} \geq c\sigma^2 \log n/(n\phi_0^2)\}$, $c > c_0$: $\pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$.

Status and evidence

T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2). T02(i♯), T02(ii♯): PROVED; SF/XF VERIFIED. T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY [SoP/04_theory.tex:765]. Statement [SoP/04_theory.tex:774-782]. Algorithm-free.

Scope. "No expectation rate inferred from O_P, no universal onset n_0, no random-design or growing-sparsity extension" (card Repairs row); $p \leq n^A$ "comes from the prior and proof technique; e^{Cn^α} OPEN". The binding condition of the exhibited families first holds near $n \approx 2 \cdot 10^6$ — an illustration for those families, not a universal onset [SoP/04_theory.tex:787-811].

Random designs. T03♯ is a fixed-design theorem. Random-design versions used below: (a) W1-PXN (P) on LH-iid, $1-\pi(S_0) = O_P(\sqrt{\log n/n})$ under the joint law of (X^r, ε), via T03♯ along design sequences plus a design event and de-randomisation (XF VERIFIED, R19) [W1/W1-PXN/REPORT.md:49]; (b) GEN's random-design class (C4): T03♯ along every deterministic sequence in the X-events with common constants and envelopes, plus the diagonal tightness transfer (instance I-c, [W1C/C-PACK/lemmas/I4_SL_corollaries.md:57-60]). Conditional convergence alone would not suffice; the every-sequence/common-envelope quantifiers are needed.

3.2 Sure screening, HPM, MPM, thresholded models (C-PACK item 4)

Lemma 3.2 (SL0, deterministic; every data set, every n)**.** *Let* $\delta := 1 - \pi(S_0)$*, with* $\pi(S_0) := 0$ *if* $s_0 > K$*; every maximum of nonnegative coordinate errors over an empty set is defined as zero. Then: (i)* $|\mathrm{PIP}_j - 1\{j \in S_0\}| \leq \delta$ *for every* j*, hence* $\max_{j\in S_0}(1-\mathrm{PIP}_j) + \max_{j\notin S_0} \mathrm{PIP}_j \leq 2\delta$*; (ii)* $|\pi(A \subseteq S) - 1\{A \subseteq S_0\}| \leq \delta$ *for every* A*, and if* $S_0 \in \Omega$*,* $|\pi(h) - h(S_0)| \leq \delta\,\mathrm{osc}(h)$ *and* $\sup_{h:\Omega\to[0,1]} |\pi(h) - h(S_0)| = \delta$*; (iii)* $\delta < \min(q, 1-q) \Rightarrow \xi_q = S_0$*, in particular* $\delta < \frac{1}{2} \Rightarrow \mathrm{MPM} = S_0$*; (iv)* $\delta < \frac{1}{2} \Rightarrow S_0$ *is the unique maximiser of* π*; (v)* $\pi(|S| = s_0) \geq 1 - \delta$*. (vi) When* $1 \leq s_0 \leq K$ *and* $s_0 < p$*, the constant 2 in (i) is sharp over arbitrary laws on* Ω *(two-point law); attainment by a D0 posterior is* not *claimed.*

Corollary 3.3 (SL1)**.** *If* $\delta_n = O_P(r_n)$*,* $r_n \to 0$*, under some law: (a)* $\max_j |\mathrm{PIP}_j - 1\{j \in S_0\}| = O_P(r_n)$ *and* $\mathbb{E}\max_j |\cdot| \leq \mathbb{E}\delta_n \to 0$ *(no expectation rate); (b) for every deterministic sequence* $q_n \in (0,1)$*,* $\mathbb{P}(\xi_{q_n} = S_0) \to 1$ *whenever* $r_n = o(\min(q_n, 1-q_n))$*, in particular for every fixed* $q \in (0,1)$*, and* $\mathbb{P}(\mathrm{MPM} = S_0) \to 1$*; (c)* $\mathbb{P}(\mathrm{HPM} = S_0) \geq \mathbb{P}(\delta_n < \frac{1}{2}) \to 1$*,* $\mathbb{P}(\mathrm{HPM} \neq S_0) \leq 2\mathbb{E}\delta_n$*; (d)* $\sup_{h:\Omega\to[0,1]} |\pi(h) - h(S_0)| = O_P(r_n)$*; (e)* $\pi(|S| = s_0) = 1 - O_P(r_n)$*. Instances with* $r_n = \sqrt{\log n/n}$*:* ***I-a*** *all T03♯ hypotheses (fixed design);* ***I-b*** *all W1-PXN (P) hypotheses (LH-iid, joint law);* ***I-c*** *GEN (C4) (joint law);* ***I-d*** *LH exact Gram (Thm5-LH keeps SAME-FAMILY-ONLY).*

Corollary 3.4 (SL1′: no beta-min, large signals only)**.** *Retaining T03♯'s other hypotheses,* $\max_{j\in J_n(c)}(1 - \mathrm{PIP}_j) \leq \pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$ *(*$= 0$ *if* $J_n(c) = \emptyset$*). No specificity or full-support recovery follows from this clause alone.*

Status and evidence

SL0, SL1, SL1′: PROVED (by this seat, unchecked), independently re-derived by R34; v2 re-read by R38 (mid-point CONTINUE) and R39 (sweep); C-PACK council round 1 passed (Sol 9.0, Opus 8.7) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:3-7]. Statements [W1C/C-PACK/lemmas/I4_SL_corollaries.md:20-48]; [W1C/C-PACK/lemmas/I4_SL_corollaries.md:53-68]. Falsification: 216 exact D0 posteriors, 84,096 pairs, 0 violations (W1C/C-PACK/numerics/R34_falsify_out.txt).

3.3 Misspecified candidate sets (C-MIS; Song–Liang Thm 4 / Cor 2 analog)

Setting. A candidate set $\mathcal{C}$, $|\mathcal{C}| = p_{\mathcal{C}} \le n^A$; the method is *re-run on the candidate data* $(X^r_{\mathcal{C}}, y^r)$ with the shipped score and prior $(\mathrm{BB}(1, p_{\mathcal{C}}), K = \min(p_{\mathcal{C}}, \lfloor n/2 \rfloor))$ — not obtained by conditioning the full-p posterior. True model $y^r = \alpha\mathbf{1} + X^r_{\mathcal{C}}\beta_{\mathcal{C}} + \omega + \varepsilon$, $\omega = X^r_{\mathcal{M}}\beta_{\mathcal{M}}$ the omitted contribution [W1C/C-MIS/STATEMENT.md:37-50].

(GR) Gaussian residual. Conditionally on $X^r_{\mathcal{C}}$, $\omega = a\mathbf{1} + X^r_{\mathcal{C}}\gamma + \xi$ with $\xi \sim N_n(0, \tau^2 I_n)$, (a, γ) measurable in $X^r_{\mathcal{C}}$, $\tau^2 \ge 0$ **fixed**, ξ independent of ε; the method's uniforms independent of everything. Define $\beta^{\mathcal{C}} := \beta_{\mathcal{C}} + \gamma$, $S_0^{\mathcal{C}} := \operatorname{supp} \beta^{\mathcal{C}}$, $s_0^{\mathcal{C}} := |S_0^{\mathcal{C}}|$, $\sigma^2_{\mathcal{C}} := \sigma^2 + \tau^2$. Under GR the candidate-only problem *is* a well-specified Gaussian linear model with truth $\beta^{\mathcal{C}}$ and noise $\sigma^2_{\mathcal{C}}$ [W1C/C-MIS/STATEMENT.md:54-58].

(H-MIS) = the T03♯ hypotheses for the candidate problem with

$$(p, S_0, s_0, \theta^*, \mu, \sigma^2) \mapsto (p_{\mathcal{C}}, S_0^{\mathcal{C}}, s_0^{\mathcal{C}}, \theta^{\mathcal{C}}, \mu^{\mathcal{C}}, \sigma^2_{\mathcal{C}}),$$

in particular $s_0^{\mathcal{C}} \ge 1$ fixed and beta-min relative to $\sigma^2_{\mathcal{C}}$.

Theorem 3.5 (MIS). *(i)* Fixed candidate design*, GR and (H-MIS):* $1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y) = O_P(\sqrt{\log n/n})$*;* $\max_j |\mathrm{PIP}_j - 1\{j \in S_0^{\mathcal{C}}\}| \le 1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)$*;* $\mathbb{P}(\mathrm{MPM}_{\mathcal{C}} \ne S_0^{\mathcal{C}}) \to 0$*; for the exact-posterior HPM* $S_{\mathcal{C}}^{\dagger} = \arg\max \pi_{\mathcal{C}}$*,* $\mathbb{P}(S_{\mathcal{C}}^{\dagger} \ne S_0^{\mathcal{C}}) \le 2\mathbb{E}[1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)] \to 0$*; without beta-min,* $\pi_{\mathcal{C}}(S \not\supseteq J_n^{\mathcal{C}}(c) \mid y) = O_P(\sqrt{\log n/n})$ *for fixed* $c > c_0$*. Consequences: a* proxy *(*$j \in S_0^{\mathcal{C}} \setminus S_0$*) has* $\mathrm{PIP}_j \to 1$*; a* cancelled *true candidate (*$\beta_j + \gamma_j = 0$*) has* $\mathrm{PIP}_j \to 0$*.* ***Corollary N****: every* $T \ne S_0^{\mathcal{C}}$ *has* $\pi_{\mathcal{C}}(T \mid y) \le 1 - \pi_{\mathcal{C}}(S_0^{\mathcal{C}} \mid y)$*; in particular the retained truth* $S_0 \cap \mathcal{C}$ *and the sample-projection support lose all mass whenever they differ from* $S_0^{\mathcal{C}}$*.*
(ii) Random candidate design, envelope form*: the same conclusions under the joint law, if on measurable design sets* $\mathcal{A}_n$ *with* $\mathbb{P}(\mathcal{A}_n) \to 1$ *every* $x \in \mathcal{A}_n$ *satisfies the finite-n inequalities of (H-MIS) with fixed constants,* $s_0^{\mathcal{C}}(x) = s \ge 1$ *and the beta-min envelope (diagonal argument, Lemma DR); (ii′) without beta-min, the inclusion clause.*
(iii) Jointly Gaussian rows *(*$\mathcal{M} := S_0 \setminus \mathcal{C}$*, rows iid* $N(m, \Sigma)$*,* $\Sigma_{\mathcal{C}\mathcal{C}} \succ 0$*): at each n GR holds with* $\gamma = \Sigma_{\mathcal{C}\mathcal{C}}^{-1}\Sigma_{\mathcal{C}\mathcal{M}}\beta_{\mathcal{M}}$*,* $\tau^2 = \beta_{\mathcal{M}}^{\top}(\Sigma_{\mathcal{M}\mathcal{M}} - \Sigma_{\mathcal{M}\mathcal{C}}\Sigma_{\mathcal{C}\mathcal{C}}^{-1}\Sigma_{\mathcal{C}\mathcal{M}})\beta_{\mathcal{M}}$*;* $\beta^{\mathcal{C}}$ *is the population slope with intercept (= Song–Liang's* β_0 *and* $S_0^{\mathcal{C}} = \tilde{s}$ *for centred rows). MIS (i)–(ii) apply only when in addition* $\tau_n^2 \equiv \tau^2$ *is fixed.* ***Neighbour rule*** *(Prop. P(d)): with one active omitted variable k and* Θ *the precision of the marginal vector* $(x_{\mathcal{C}}, x_k)$*, a candidate that is null in the full model belongs to* $S_0^{\mathcal{C}}$ *iff* $\Theta_{jk} \ne 0$ *— partial, not marginal, correlation decides (AR(1): the two neighbours; a one-factor design can make* $s_0^{\mathcal{C}}$ *grow, outside the theorem).* ***Lemma GD****: (a) for fixed* $p_{\mathcal{C}}, m, \Sigma, \beta$ *fixed,* $\beta^{\mathcal{C}} \ne 0$*, (ii) applies and the inclusion clause recovers Song–Liang's Cor. 2 conclusion in the jointly Gaussian case, now with a rate; (b) growing* $p_{\mathcal{C}} \le n^A$ *under stated population restricted-eigenvalue, variance, energy and projected beta-min assumptions — projection sparsity is* assumed*.*

Theorem 3.6 (F: fixed design, exact sparse projection)**.** *Put* $N = n - 1$*; let* $U^{\top}U = I_N$*,* $UU^{\top} = C$*,* $\mu_c = U^{\top}X^r\beta$ *(the full mean in contrast coordinates) and* $X_{c,\mathcal{C}} = U^{\top}X_{\mathcal{C}}$*. Here* $R^{\mathcal{C}}$ *means* $\|\mu_c\|^2 \le Mn$ *and* $\lambda_{\min}(\hat{R}_{S_0^{\mathcal{C}}}) \ge \phi_0^2$ *for fixed constants; the energy bound is on the full mean. Assume (EP):* $r := \mu_c - X_{c,\mathcal{C}}\theta^{\mathcal{C}}$ *is orthogonal to every candidate,* $\operatorname{supp} \theta^{\mathcal{C}} = S_0^{\mathcal{C}}$*; set* $\Delta_n = \|r\|^2/N$*,* $\sigma^2_{\mathcal{C},n} = \sigma^2 + \Delta_n$*. Under* $s_0^{\mathcal{C}} \ge 1$ *fixed,* $p_{\mathcal{C}} \le n^A$*, G,* $R^{\mathcal{C}}$*,* $SE_\lambda(j_0)$*,* $SE_w(m_n; g_0)$ *anchored at* $S_0^{\mathcal{C}}$ *(cutoff* $s_0^{\mathcal{C}} \le m_n \le \min(K, C\log n)$*), BM-*$\infty$ *or BM(*$c > c_0$*) relative to* $\sigma^2_{\mathcal{C},n}$*, and the size tail* $\boldsymbol{TAIL}^{\mathcal{C}}(m_n)$*:* $\pi_{\mathcal{C}}(|S| > m_n \mid y) = O_P(\sqrt{\log n/n})$ *— the conclusions of T03♯ hold for* $S_0^{\mathcal{C}}$*.* $TAIL^{\mathcal{C}}$ *may be removed (a) for bounded* $p_{\mathcal{C}}$*, taking* $m_n = m_n^{\sharp} = K$ *eventually (Cor. F(a)); or (b) if, for a fixed* $\kappa \in (0, 1)$*, (TL)* $\sigma^2/[\Delta_n + \sigma^2(1 - d_n/N)] \le \frac{3}{4}(1 - \kappa)$ *eventually,* $d_n = \operatorname{rank} X_{c,\mathcal{C}}$*. In (b), Lemma T supplies the deterministic* $m_n^T = s + \lceil (s(A + \frac{1}{2}) + 1)\log n/\log(1/\rho) \rceil$ *(*$s = s_0^{\mathcal{C}}$*,* $\rho = \frac{1}{2}(\frac{1}{4} + \frac{3\kappa}{8})^{-1/2}$*) with* $\pi_{\mathcal{C}}(|S| > m_n^T \mid y) = O_P((\log n)^s/n)$*; take* $m_n = \min(K, \max(m_n^{\sharp}, m_n^T))$ *and require* $SE_w(m_n; g_0)$ *at this cutoff; all other hypotheses of Theorem F remain (Cor. F(b))* [W1C/C-MIS/lemmas/L4_theoremF.md:158-162]*.* ***Lemma Z*** *(null projection, fixed candidate design): if* $s_0^{\mathcal{C}} = 0$*,* $p_{\mathcal{C}}$ *is bounded, and* $\lambda_{\min}(X_{\mathcal{C}}^{\top}X_{\mathcal{C}}) \ge g_0 > 0$ *eventually, then under GR, or under EP with* $\theta^{\mathcal{C}} = 0$*,* $1 - \pi_{\mathcal{C}}(\emptyset \mid y) = O_P(n^{-1/2})$*. The general growing-p null case is not claimed.*

Status and evidence

C-MIS: PROVED-CHECKED (narrowed form): Theorem MIS (i)–(iii) with Prop. P, Lemma DR, Lemma GD(a)/(b); Corollaries H, D, D′; Theorem F with F(a), F(b), Lemma T; Lemma Z; Corollary N; Examples E1–E3. Statement lock R01; lemma checks R02, R03, R05; mid-point R04 CONTINUE; sweep R06 (0 fatal); council round 1 Sol 9.2 / Opus 9.0, all gates PASS [W1C/C-MIS/REPORT.md:21-29]. Statements [W1C/C-MIS/STATEMENT.md:64-95]; [W1C/C-MIS/STATEMENT.md:126-141]; [W1C/C-MIS/STATEMENT.md:101-105]. Inputs: T03♯ as above; T38-(0) PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5)).

Boundary. Example E1: x_1, x_2, x_3, z iid $N(0,1)$, omitted $x_k = -x_1 + x_2 + z$, $y = x_1 + x_k + \varepsilon$: $S_0^{\mathcal{C}} = \{2\}$ (proxy), the true candidate 1 is cancelled, and both $\{1\}$ and the dense sample-projection support lose their mass. E2: a common random omitted slope with Gaussian marginals defeats the transfer (GR's covariance content matters). E3: EP cannot be dropped for a pre-named target. **Open** (§8): EP with growing $p_{\mathcal{C}}$ and small residual $0 < \Delta_n \lesssim \sigma^2(1/3 + d_n/N)$; approximate projection; outside GR; random-design EP [W1C/C-MIS/REPORT.md:131-139].

4 Sampler level

4.1 Unbiasedness of every bounded summary per measured round (M1(b), Lemma N1)

Throughout this subsection, fix an admissible standardised data set (X, y), with $n \geq 3$ and $p \geq 1$. All expectations and almost-sure assertions in Lemma N1 and Corollary E are over the algorithm's streams only; Z, π and S_0 are fixed. If these identities are embedded in the joint sampling law, condition additionally on (X, y), retaining the same localisation on E_r [W1/W1-N1/STATEMENT.md:62-66].

Island r runs on final stream $9 + r$ after a calibration record $\mathcal{C}$ frozen on streams 0–9; $\hat{Z}_r$ is its evidence estimate, Z the normalising constant, $W_r := \hat{Z}_r / Z$. After each of the four measured `population_round(P,1.0)` calls the population is $(S_{rqi})_{i \leq N_e}$, $q = 1, \ldots, 4$ [CORE/cpp/v3_engine.h:355-361]; the v3.1 accumulator is $H_r(h) = (4N_e)^{-1} \sum_{q,i} h(S_{rqi})$, which for $h = 1\{j \in S\}$ is exactly the coded `pip_plain[j]`.

Premises (named). EX-A; EX-S/H-U (iid uniforms within stream $9 + r$); START (`draw_pi0` exact, $\bar{Z}_0$ exact; N9 PROVED (candidate)); KERNEL (the coded steps realise the T12 kernels; N9 PROVED (candidate), code-checked, CHK_INHERITED VERIFIED WITH AMENDMENT); CAL; SI (stream $9 + r$ independent of streams 0–9; for $r > 4$ implied by IND-T, whose extension audit is pending); MEAS; the potential (uninterrupted) run; worker equivalence for v3.1's concurrent workers [W1/W1-N1/STATEMENT.md:68-91].

Let $\mathcal{A} \in \sigma(\mathcal{C})$ be the successful-calibration event and $E_r = \mathcal{A} \cap \{r \leq R'\}$. An identity $\mathbb{E}[Y \mid \mathcal{C}] = b$ "almost surely on E_r" means $\mathbb{E}[1_{E_r} Y \mid \mathcal{C}] = 1_{E_r} b$ almost surely; no island record is asserted outside E_r [W1/W1-N1/STATEMENT.md:94-99].

Lemma 4.1 (N1). *For every island r, every $q \in \{1, \ldots, 4\}$ and every jointly measurable $h(S; \mathcal{C})$ with a deterministic bound $|h| \leq B_h$, fixed before stream 10, almost surely on $\mathcal{A} \cap \{r \leq R'\}$ (localised form):*

$$\mathbb{E}\Big[\hat{Z}_r \cdot \tfrac{1}{N_e} \textstyle\sum_i h(S_{rqi}) \;\Big|\; \mathcal{C}\Big] = Z\,\pi(h), \qquad \textit{hence} \qquad \mathbb{E}[\hat{Z}_r H_r(h) \mid \mathcal{C}] = Z\,\pi(h).$$

No regime assumption on X is made.

Corollary 4.2 (E = M1(e), constant δ_1). *Under N1's named premises, almost surely on E_r in the same localised sense: for $h \in [0,1]$ (as in Lemma N1, defined on $\Omega \cup \{S_0\}$) and $\delta_1 := 1 - \pi(S_0)$: $\mathbb{E}[W_r | H_r(h) - h(S_0)| \mid \mathcal{C}] \leq \mathbb{E}[W_r H_r(1\{S \neq S_0\}) \mid \mathcal{C}] = \delta_1 \leq 2\delta_1$, attained by $h = 1\{S \neq S_0\}$.*

Status and evidence

W1-N1: PROVED-CHECKED; N1 PROVED under the named premises — cross-family checked, council-passed (round 1); Corollary E PROVED under the same premises; council Codex 9.3 / Opus 8.9 [W1/W1-N1/REPORT.

md:44-46]. Statement [W1/W1-N1/STATEMENT.md:95-111]. Stronger than the card in three places (any bounded h; each measured round; constant δ_1). **Not Lean-certified** (Lean not supplied; HQ to waive or open a registrar task). The SoP card still reads N1: PROOF SKETCH [SoP/04_theory.tex:123]; the registry update is HQ's.

4.2 What holds for the reported PIPs

The paper must distinguish four PIP objects [W1C/C-PACK/lemmas/I4_SL_corollaries.md:70-88]:

1. **Exact PIPs** $\mathrm{PIP}_j = \pi(j \in S)$: consistent with rate (Cor. 3.3).
2. **Plain measured-endpoint PIP, islandwise** (shipped $R' = 4$): Corollary 4.2 with $h = 1\{j \in S\}$, evidence-weighted, in expectation given $\mathcal{C}$, for the potential run. This is *not* the study's `pip_primary` and not a statement in probability about the pooled estimate.
3. **Pooled reported PIPs.** The cited cards contain **no in-probability sure-screening theorem for the pooled shipped-$R' = 4$ estimator**. M1(i), status (h), (i): PROVED (by this seat, unchecked) [SoP/04_theory.tex:130], gives $\hat{\mu}(h) \to h(S_0)$ in probability jointly with output for a *fixed finite* declared list under T03♯'s hypotheses, $\operatorname{osc} d \le D_* = O_P(1)$ and the stacking configuration $R'_n = 4\lceil \log_2(n+p+1) \rceil$; simultaneous control of $\le (n+p)^2$ declared summaries needs the card's separate configuration $R'_n = 4\lceil (n+p)^6/4 \rceil$ [SoP/04_theory.tex:111]. The finite-R' bound M1(f) is "vacuous at $R' = 4$" [SoP/04_theory.tex:117]. With N1 now proved (Lemma 4.1), the premise "E[W_r $\hat{h}_r$] = π(h)" of PC1c3 used by M1(f),(g),(i) is discharged islandwise; those clauses keep their own statuses and remaining premises (IND-T extension for $R' > 4$, GUARD-W, stacking, T03♯) [W1/W1-N1/REPORT.md:48].
4. **Store-restricted PIPs** $\mathrm{PIP}^{\mathcal{T}}$ (π conditioned on the reported store $\mathcal{T}$), Corollary SL2 below.

Corollary 4.3 (SL2: store-restricted PIPs, jointly with output)**.** *Assume M4(b) (if $S_0 \in \mathcal{T}$: $|\mathrm{PIP}^{\mathcal{T}}_j - \mathrm{PIP}_j| \le \delta_1$), M4(c) (if $S_0 \in \mathcal{S}^+$ and $\delta_1 < \frac{1}{2}$ then $S_0 \in \mathcal{T}$, under LS0 with $\mathcal{T}_{\mathrm{new}} = \mathrm{top}_c(\mathcal{T}_{\mathrm{old}} \cup \mathcal{S}_{\mathrm{added}})$), exact score ordering, a positive-capacity store, truth discovery with output $\mathbb{P}(\mathcal{O} \cap \{S_0 \notin \mathcal{S}^+\}) \to 0$ (PXN or GEN), and $\delta_n = O_P(r_n)$. Then $\mathbb{P}(\mathcal{O} \cap \{\max_j |\mathrm{PIP}^{\mathcal{T}}_j - 1\{j \in S_0\}| > 2\delta_n\}) \to 0$, and the same with any deterministic threshold t_n, $r_n = o(t_n)$.*

Status and evidence

SL2: PROVED (by this seat, unchecked), R34 re-derived; C-PACK council passed. Inputs verbatim: M4(a) MZ-P3: "PROVED (P3 top store, deterministic); P3' conditional on T03 + discovery"; "candidate: not reviewed, not adopted"; AUTHOR-ONLY; M4(b) same as (a); W1PC1 §3 B.D3(b) COUNTEREXAMPLE repaired with $\|h\|_\infty$; M4(c) N13: PROVED (PC1 claim 5) [W1C/C-PACK/lemmas/I4_SL_corollaries.md:112-114]. Statement [W1C/C-PACK/lemmas/I4_SL_corollaries.md:90-103].

Not claimed / not proved — sampler level

- No in-probability sure-screening or selection theorem for the shipped pooled $R' = 4$ PIPs or MPM; no TV/KL bound for the final pooled estimate under masking (CJ02).
- N1 and Corollary E are about the potential run; delivered records equal potential ones on $\mathcal{O}$ only through GUARD-W (audit open). Worker equivalence, the IND-T extension ($R' > 4$) and START/KERNEL (N9, candidate) remain premises.

5 Reported-HPM selection consistency

5.1 The route: M5(0), operational closure, LS0

By M5(0) (§1) it suffices to show $\mathbb{E}_0[1-\pi(S_0)] \to 0$ (layer (i), §3) and $D_n \to 0$. LS0 reads no uniforms, so the models it scores are a deterministic function of the data; under the named premises the **operational closure** [OP] gives

$$\bar{\mathcal{O}} \cap \{S_0 \notin \bar{S}^+\} \subseteq \{\text{LS0 from } \tilde{S}_0 \text{ does not score } S_0\}, \qquad \text{so} \qquad D_n \le \mathbb{P}(\text{LS0 misses } S_0)$$

(PROVED (by this seat, unchecked) inside Theorem GEN PROVED-CHECKED; XF VERIFIED as a theorem under (OP) in W1-PXN) [W1/W1-GEN/STATEMENT.md:169-171]; [W1/W1-PXN/REPORT.md:43]. Every theorem in this section is therefore a statement about two data events: the start mode lies in a region from which the greedy search reaches S_0, and the search does reach it within its move cap. **None of them is a delivery claim**; the liminf form $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$ is algebra from $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$.

5.2 The start-mode centre: e4 is not needed (C-PACK item 1)

Lemma 5.1 (UM). *Under W1-N3 (H1)–(H4) (correct encoding of the* `Start` *tables, exact arithmetic, enumeration cap* ≥ 1*, C++ execution limits), with* $f(S) = \log w(|S|) + \sum_b \log r_b(S \cap C_b)$ *(the log start weight, up to a model-independent constant), if* $\arg\max_\Omega f = \{R\}$ *then* `start_mode(st, cap).S` $= R$ *(from N3(c) with* $N = 1$*, or N3(a)); the truncation flag is irrelevant. In particular this holds on* $\{\pi_0(R) > \frac{1}{2}\}$.

The positive consistency proofs of PXN, GEN and M5(b1-LH) use the LS0 centre only to identify it with a deterministic reference $R \in \{S^\circ, S^*\}$ on $\{\pi_0(R) > \frac{1}{2}\}$; hence premise e4 is replaced by **e4′**: (i) the LS0 centre is `start_mode(st, cap).S` of the coded routine, or a replacement returning the unique maximiser whenever it is unique (C-PACK MR1); (ii) the routine uses the same retained indices, K, ordered actual partition, full-response tables and prior as the mathematical π_0, and the coded centre is a deterministic measurable data function on every branch; (iii) wherever LS0 is required its emitted centre is this function, with OP-LS0/scored-set/guard contracts unchanged; (iv) the consumer's EX-A. The coded-routine option retains W1-N3's encoding/execution conditions H1 and H4 and an enumeration cap at least one (fixed in advance or a specified deterministic measurable data function satisfying H3–H4); this does not require the lexicographic tie fallback for the positive consistency proofs [W1C/C-PACK/lemmas/I1_e4_replacement.md:141-160]. The lexicographic tie rule and the truncation fallback are needed by no positive consistency statement; Theorem GROW needs the emitted centre to be "some maximiser" of γ_0 also when ties exist — which e4′(i)'s "unique-maximiser" replacement clause does not guarantee, but the coded routine (W1-N3(a): always a smallest-size maximiser) and TDP do [W1C/C-GROW/lemmas/00_MAIN.md:31-33]. A binary64 version (Lemma UM-FP, under FP-TAB $\Delta(R) > 2(\eta+\varepsilon)$) covers the start-mode identification step only; FP-TAB is untested for the coded scorer.

Status and evidence

C-PACK items 1–4: PROVED-CHECKED (council round 1: Codex gpt-6.1-sol xhigh 9.0, Claude Opus 5.5 high 8.7, all six gates PASS) [W1C/C-PACK/REPORT.md:19-21]; item 1 PROVED; XF-checked (R33 GAP → 7 fixes), re-read R38/R39, council. W1-N3 (the code's actual start-mode behaviour; counterexample above the cap 10^6; reference algorithm TDP): PROVED-CHECKED (council Sol 9.3 / Opus 8.6); "e4 for integrated LS0 stays OPEN" as an integration item [W1/W1-N3/REPORT.md:12-14].

5.3 Theorem PXN: masked correlated design, p ≫ n (LH-iid)

Family LH-iid. Fixed $g \ge 2$, $c \ge 1$, $v \ge 0$, $\rho \in (0,1)$, links $0 < |\psi_i| < \rho$, nonzero $\beta_i, \beta_h^{\text{vis}}$, $\sigma^2 > 0$, $A \ge 1$. $p_n = G_n g$, $18 \le p_n \le n^A$, $p_n \to \infty$. Rows iid $N(0,\Sigma)$: groups of size g equicorrelated

at ρ; c linked group pairs with rank-one cross-correlation through the hubs ($\mathrm{corr}(j_a, j_b) = \psi$); v visible groups; the rest null. Truth: $\theta_{j_{a_i}} = \beta_i$, $\theta_{j_{b_i}} = -\psi_i\beta_i$, visible hubs β_h^{vis}; $s_0 = 2c + v$. The b-hubs are *masked* (zero population covariance of their block with the signal); reference $S^\circ = S_0 \setminus \mathcal{M}$. **(SIZE)** $g \le B_n < 2g$ eventually ($9 \le g \le 16$ asymptotically) [W1C/C-PACK/item5_PXN_consolidated/00_statement_and_conventions.md:8-30]. Method: v3 + LS0 (one production local search from the exact start mode, centre emitted, cap ℓ_n), under (OP) = EX-A, EX-S, CAL, SR, GUARD, OP-LS0, with the **exact-branch convention** (blocks, tables, π_0, $\tilde{S}_0$ and "LS0 of $\bar{R}^{+}$" are the deterministic data functions on every branch).

Write $s = |S^\circ| = c + v$, $\bar{\psi} = \max_i |\psi_i|$, $q_R = \sqrt{(1-\bar{\psi}^2)/(1-\rho^2)}$, $t^* = (q_R - 1)/(q_R\rho - \bar{\psi}) \in (\bar{\psi}, \rho)$ and $I(t; r) = -\frac{1}{2}\log[(1-r^2)(1-t^2)/(1-rt)^2]$.

Theorem 5.2 (PXN). *Under LH-iid with (SIZE), $18 \le p_n \le n^A$, $p_n \to \infty$, s_0 fixed, and (OP), as $n \to \infty$:*

(R) $\mathbb{P}(\textit{blocks} \neq \textit{groups}) \to 0$, *summable, and for every fixed $C > 0$, $\le p^{-C}$ for $n \ge \max\{n_1, n_0(A, C, \rho, \bar{\psi})\}$ (n_1 the SIZE onset); explicitly*

$$\mathbb{P}(\textit{blocks} \neq \textit{groups}) \le \frac{2}{\sqrt{\pi}(1-\rho)} p^2 \sqrt{n}\, e^{-(n-4)I^*}$$

for $n \ge 5$ with SIZE at n, $I^* = I(t^*; \bar{\psi}) > 0$.

(S) $1 - \pi_0(S^\circ) = O_P(\sqrt{\log n/n})$; $\pi_0(S_0) = p^{-c} n^{-c/2} e^{O_P(1)} \to 0$, *more precisely* $\log \pi_0(S_0) + c\log p + \frac{c}{2}\log n \to_d \sum_{i \le c} \log \frac{s+i}{2} + \frac{1}{2}\chi^2_c$ *(the masked regime).*

(E) $\mathbb{P}(\tilde{S}_0 = S^\circ) \to 1$.

(T) $\mathbb{P}(\textit{LS0 scores } S_0) \to 1$ *and* $\mathbb{P}(\textit{LS0 returns } S_0 \textit{ with } \texttt{local_certified}) \to 1$.

(P) $1 - \pi(S_0) = O_P(\sqrt{\log n/n})$ *under the joint law of (X^r, ε).*

(C) For every n: $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \le 2\mathbb{E}_0[1 - \pi(S_0)] + D_n$, $D_n \le \mathbb{P}(\textit{LS0 misses } S_0) \to 0$; *hence* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$, $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) - (1 - \mathbb{P}(U_n)) \to 0$, $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$, *and* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) \to 1 - u$ *whenever* $\mathbb{P}(U_n) \to u$.

(K) LS0 costs $\le 1 + (\ell_n + 1)\max_{k \le K}[p + k(p-k)] = O(p^2 \log(n+p))$ *scores; on the joint event $\mathcal{T}_n$ (lattice arrival, (L), $\ell_n \ge c$) it uses $c + 1$ scans.*

Corrected centring remark. The finite-n centring added after the W1-PXN council claimed a remainder $O_P(\sqrt{\log n/n})$; this fails for slowly growing p. The correct remainder is $O_P(\sqrt{\log n/n} + 1/p)$ (prior correction $d_p = \sum_{i \le c} \log[2p/(2p - s - i)] = O(1/p)$); the limit law in (S) and the headline are unaffected, and no rate of convergence to the χ^2 law is asserted [W1C/C-PACK/item5_PXN_consolidated/03_start.md:111].

Status and evidence

W1-PXN: PROVED-CHECKED (council round 1: Codex 8.6, Opus 8.3, all gates PASS) [W1/W1-PXN/REPORT.md:3-6]. Theorem as stated in the consolidated supplement [W1C/C-PACK/item5_PXN_consolidated/00_statement_and_conventions.md:97-119]; supplement fidelity FAITHFUL-WITH-FIXES (R36), all fixes applied. C-PACK addenda (separately identified, not silently merged): e4 → e4′ and Lemma T(i)'s off-event centre identity replaced by $\mathbb{P}(C_0 \neq S^\circ) \le \mathbb{P}(\mathcal{A}_n^c) + \mathbb{P}(\tilde{\pi}_0(S^\circ) \le \frac{1}{2})$. **A-dependence**: no step uses $p \le n$; A enters only constants and onsets; a sufficient onset envelope is $n/(\log n)^3 \ge C(A, \text{fixed parameters})$ plus SIZE; no uniformity over growing A.

5.4 Theorem GEN: a general correlated class with a bounded relevance set

Fixed-design model G, standardised design, $s_0 \ge 1$ fixed, $p \le n^A$; method v3.1 + LS0 as specified with cap ℓ_n; premises EX-A, EX-S, CAL, SR, GUARD-W, OP-LS0, e4 (read e4′), complete scans; IND-T and the

c_4 search when $R' > 4$. Notation: $\Lambda_T = \|(I - P_T)\mu\|^2$, $\Delta_T = \Lambda_T/(n-1)$, $v_T = \sigma^2 + \Delta_T$, $\delta_{T,l} = u_{T,l}^{\top}\mu$ with $u_{T,l}$ the normalised residual of x_l on X_T.

- **(H0)** the T03♯ hypotheses with BM-∞.
- **(H1) bounded separated relevance set**: deterministic $\mathcal{V}_n \supseteq S_0$, $|\mathcal{V}_n| \le \bar{v}$: (H1a) $\lambda_{\min}(\hat{R}_{\mathcal{V}\cup\{l\}}) \ge \phi_0^2$ for every l; (H1b) SEP(c_S): $\max\{\Lambda_S - \Lambda_{S\cup\{l\}} : S \subseteq \mathcal{V}, l \notin \mathcal{V}\} \le c_S(n-1)\phi_0^4\theta_{\min}^{*2}$ eventually, fixed $c_S \in [0,1)$.
- **(H2) centre certificate on the actual blocks**: deterministic $S^* \subseteq \mathcal{V}$, $|S^*| \le \bar{v}$, $U_b^* = S^* \cap C_b$: (H2a) within-block SE: $\lambda_{\min}(\hat{R}_T) \ge \phi_0^2$ for every nonempty $T \subseteq C_b$, every b; (H2b) VIS: $\|(P_{U\cup U_b^*} - P_U)\mu\|^2 \ge M_n v_U \log n$ for every b and every $U \subseteq C_b$ with $U_b^* \not\subseteq U$, for a deterministic $M_n \to \infty$; (H2c) NL: with the leakage family $\mathcal{L}(S^*) = \{(b,T,l) : l \in C_b \setminus U_b^*,\ U_b^* \subseteq T \subseteq C_b \setminus \{l\}\}$ and fixed $C_L < \infty$, $\kappa > 0$, $\varepsilon_0 > 0$: $\delta_{T,l}^2 \le C_L\sigma^2 \log n$ on $\mathcal{L}(S^*)$ and, if $p > s^*$, eventually $(p-s^*)^{-1}\sum_{l\notin S^*} \max_{T:(b(l),T,l)\in\mathcal{L}(S^*)} \exp(E_{T,l}) \le n^{1/2-\kappa}$, where $h_n = \sqrt{2(A+\varepsilon_0)\log n}$ and $E_{T,l} = \min\{\delta_{T,l}^2/(2\Delta_T),\ \delta_{T,l}^2/(2v_T) + |\delta_{T,l}|\sigma h_n/v_T\}$, with $E_{T,l} = 0$ when $\delta_{T,l} = 0$ (in particular when $\Delta_T = 0$); the average is omitted when $p = s^*$ ($\mathrm{NL}_{\mathrm{avg}}(\kappa)$). Convention for (H1b): the maximum is 0 if $\mathcal{V} = [p]$ [W1/W1-GEN/STATEMENT.md:56-75].

Canonical certificate (C1′). Under (H2a) and $\|\mu\|^2 \le Mn$, let γ_b be the coefficients of the projection of μ on the block's own columns. If $\gamma_b = \gamma_b^L + \gamma_b^S$ (disjoint supports) with $\|X_{C_b}\gamma_b^S\|^2 \le \eta\sigma^2 \log n$ and $\sqrt{(n-1)\phi_0^2}\min|\gamma_{b,j}^L| \ge \sqrt{\eta\sigma^2\log n} + \sqrt{M_n(\sigma^2+2M)\log n}$ whenever $\operatorname{supp}\gamma_b^L \ne \emptyset$, for a fixed $\eta \in [0, (\sqrt{2A+1} - \sqrt{2A})^2)$ and some $M_n \to \infty$, then $S^* := \bigcup_b \operatorname{supp}\gamma_b^L$ satisfies (H2b) and $\mathrm{NL}_\eta(S^*)$ [W1/W1-GEN/STATEMENT.md:89-97]; **amendment P-L8.1**: (H2c) is then supplied with $C_L = \eta$ and $\varepsilon_0 > 0$ small enough that $\eta/2 + \sqrt{2\eta(A+\varepsilon_0)} < \frac{1}{2}$, and any fixed $0 < \kappa \le \frac{1}{2} - \eta/2 - \sqrt{2\eta(A+\varepsilon_0)}$ ($\eta = 0$: any $\varepsilon_0 > 0$, $0 < \kappa \le \frac{1}{2}$) [W1C/C-PACK/patches/P_L8_certificate.md:7-13]. Masked truths are truths with zero (or negligible) block coefficient; they are recovered by the descent. These are oracle conditions in $(X, \beta, \sigma^2, \text{partition})$, not an observable test.

Theorem 5.3 (GEN). *(C1) Under G, $\|\mu\|^2 \le Mn$, $p \le n^A$, (H2), $s^* := |S^*| \le \bar{v}$: $1 - \pi_0(S^*) \to 0$ in probability and $\mathbb{P}(\tilde{S}_0 = S^*) \to 1$. (C2) Under G, R, BM-∞, (H1): on events of probability → 1, simultaneously for every centre $C \subseteq \mathcal{V}$, the uncapped greedy path keeps its centres in $2^{\mathcal{V}}$, reaches S_0 in $\tau(C) \le 2^{\bar{v}} - 1$ moves, and S_0 has no strictly improving feasible neighbour. (C3) Under (H0), (H1), (H2) and the premises, for $n \ge n_0$ with $\ell_n \ge 2^{\bar{v}} - 2$: $D_n \le \mathbb{P}(\tilde{S}_0 \ne S^*) + O(1/\log n) \to 0$, hence $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \to 0$ and $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$. (C3-fav) If $S^* = S_0$, (H1) is not needed. (C4) Random designs: let X^r be independent of ε, with β, $\mathcal{V}_n$ and S_n^* allowed to be X-measurable. If every deterministic sequence in measurable design sets $\mathcal{A}_n$ with $\mathbb{P}(\mathcal{A}_n) \to 1$ satisfies (H0)–(H2), respectively (H0), (H2) with $S^* = S_0$, with the same fixed constants and the same deterministic beta-min and VIS envelopes, then the conclusions of (C3), respectively (C3-fav), hold under the joint law* [W1/W1-GEN/STATEMENT.md:115-120].

Instances (C5). (a) LH with exact Gram, SIZE: $S^* = S_0 \setminus \mathcal{M}$, $\mathcal{V}$ = signal units (input: blocks = groups, W1-EXP Lemma 4.2 via Thm5-LH, SAME-FAMILY-ONLY). (b) **LH-iid, now without candidate inputs**: partition recovery and the correlation bound come from W1-PXN's repaired Theorem R (or route R′) and Lemma CR/[CONC′] ($\delta_n = 32\sqrt{2(A+1)\log n/(n-1)}$, operator error $O((\log n)^{3/2}n^{-1/2})$), with the exact-branch convention; GEN's other hypotheses, envelopes and premises are retained (C-PACK Lemma I2; "unconditional" means removal of the two candidate inputs only; onset ≈ 10^{10} for typical constants) [W1C/C-PACK/patches/P_GEN_LHiid.md:18-24]; [W1C/C-PACK/lemmas/I2_LHiid_unconditional.md:80]. (c) W (exact Gram): favourable, $S^* = S_0$. (d) **Study generator**, asymptotic sequence: groups of 5, $p_n = 5G_n \to \infty$, $p_n \le n^A$, $s_0 = 14$ in one of the study's two high-p layouts, a fixed scale $c > 0$ (amendment P-SEL.2), iid $N(0, \Sigma_s)$ rows: **S1, S3, S4, S6 covered** by (C3-fav) via (C4) (Lemma SEL); S2 and S5 outside the class. (C6) The bounded-size conditions cannot be dropped from the LS0 route: an explicit design satisfying every hypothesis except $|\mathcal{V}_n|, s^* \le \bar{v}$ has $\mathbb{P}(\text{LS0 misses } S_0) \to 1$ while $\pi(S_0) \to 1$ — a counterexample for the *LS0 route with cap ℓ_n*, not for the full method.

Status and evidence

W1-GEN: PROVED-CHECKED (council round 1: Sol 9.0, Opus 8.9, all six gates PASS) [W1/W1-GEN/REPORT.md:17-25]; statement v2.3 [W1/W1-GEN/STATEMENT.md:84-154]. C-PACK item 2 (LH-iid without candidate inputs) PROVED; XF-checked (R35 V-W-A, 3 fixes), R38/R39, council; item 3: Lemma SEL and L8 (C1′) each VERIFIED-WITH-AMENDMENTS by a dedicated xhigh check (R31, R32), amendments re-checked (R37); "No hypothesis or conclusion of Theorem GEN changes" [W1C/C-PACK/patches/P_SEL_L6d.md:42-44]. Finite n: at the study's size ($n = 200, p = 500$) the visibility margins are 0.16–0.94 (< 1); no finite-n claim.

5.5 Theorem GROW: growing relevance sets, including study S2

Same setting as GEN, but LS0 runs with the **method requirement MR1 (C-GROW)**: move cap $\ell^{\mathrm{LS0}} := \ell_n + |\tilde{S}_0|$ (all other LS0 rules unchanged; the final local search keeps ℓ_n) [W1C/C-GROW/METHOD_REQUIREMENTS.md:1-6]. For a deterministic null set $\mathcal{P}_n$ let $Q = S_0 \cup \mathcal{P}_n$ and $\mathcal{L} = \{T : S_0 \subseteq T \subseteq Q\}$.

- **(H0)** the T03♯ hypotheses with BM-∞.
- **(G1)** $|Q| \le C_Q \log n$ (C_Q fixed), $|Q| \le K$, and either $Q = [p]$ or $K \ge 2|Q| + 32s_0 + 1$, for $n \ge n_0$.
- **(G2)** $\lambda_{\min}(\hat{R}_{Q\cup\{l\}}) \ge \phi_0^2$ for every $l \in [p]$.
- **(G3)** on the actual blocks: (G3a) within-block SE: $\lambda_{\min}(\hat{R}_T) \ge \phi_0^2$ for every nonempty $T \subseteq C_b$, every b; (G3b) TV: for a deterministic $M_n \to \infty$, every $j \in S_0$ and every $U \subseteq C_{b(j)} \setminus \{j\}$: $\delta^2_{U,j} \ge M_n \log n$; (G3c) NEG($\eta$) for a fixed $\eta \in [0, (\sqrt{2A+1} - \sqrt{2A})^2)$: every $l \notin Q$ and U with $S_0 \cap C_{b(l)} \subseteq U \subseteq C_{b(l)} \setminus \{l\}$: $\delta^2_{U,l} \le \eta\sigma^2 \log n$.
- **(G4)** (G4-χ) $|\mathcal{P}_n| \le (1 - \delta_0) \log n$ for a fixed $\delta_0 \in (0, 1)$; or (G4-dir) the add directions $\{u_{T,l} : T \in \mathcal{L}, l \notin T\}$ number at most pN_n with $\log N_n = o(\log n)$.

Theorem 5.4 (GROW). *Let* $\mathrm{NA}(\mathcal{L}) = \{F(T \cup \{l\}) < F(T)$ *for all* $T \in \mathcal{L}$ *with* $|T| < K$ *and all* $l \notin T\}$ *and* $\mathrm{TL}(\mathcal{L}) = \{F(A) < F(T)$ *for every* $T \in \mathcal{L}$ *and every feasible neighbour* A *of* T *with* $S_0 \not\subseteq A\}$ *(strict inequalities). (C1) On* $NA(\mathcal{L}) \cap TL(\mathcal{L})$*, from every centre* $C \in \mathcal{L}$ *the uncapped §6.10 path deletes one element of* $C \setminus S_0$ *per move and stops at* S_0 *after* exactly $|C \setminus S_0|$ *moves; the capped search scores* S_0 *if* $\ell \ge |C \setminus S_0| - 1$ *and ends certified at* S_0 *without* `local_limit` *if* $\ell \ge |C \setminus S_0|$*. (C2) Under G,* $\|\mu\|^2 \le Mn$*,* $p \le n^A$*,* $|Q| \le C_Q \log n$*,* $|Q| \le K$*, (G2) and BM-∞:* $\mathbb{P}(\mathrm{TL}(\mathcal{L})) \to 1$*. Under the same hypotheses without BM-∞, and with (G4):* $\mathbb{P}(\mathrm{NA}(\mathcal{L})) \to 1$*. (C3) Under G,* $\|\mu\|^2 \le Mn$*,* $p \le n^A$*, (G1) and (G3):* $\mathbb{P}(S_0 \subseteq \tilde{S} \subseteq Q$ *for every maximiser* $\tilde{S}$ *of* γ_0 *on* $\Omega) \to 1$*. (C4) Under (H0), (G1)–(G4), the premises and MR1 (C-GROW):* $D_n \le \mathbb{P}($*sandwich fails*$) + \mathbb{P}(\mathrm{NA}^c) + \mathbb{P}(\mathrm{TL}^c) \to 0$*; hence* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \to 0$ *and* $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup \mathbb{P}(U_n)$*. (C4)(b) Under (G4-χ) the production cap* ℓ_n *suffices.*

Instances (C5). (a) *AR(1), exact Gram*: $X^\top X = (n-1)\Sigma$, $\Sigma_{ij} = \rho^{|i-j|}$, fixed $\rho \in (-1, 1) \setminus \{0\}$, any $p \le n - 1$ (fixed or growing), fixed s_0, fixed nonzero θ^*, fixed $\sigma^2 > 0$, any block partition with blocks of size ≤ 16, separation $\sum_{k \in S_0 \setminus \{j\}} |\theta^*_k||\rho|^{|j-k|} \le \frac{1}{2}|\theta^*_j| \frac{1-\rho^2}{1+\rho^2}$ for every $j \in S_0$. Set $d_n = \lceil \log n / (2 \log(1/|\rho|)) \rceil$, $\mathcal{P}^0_n = \{l \notin S_0 : \min_{k \in S_0} |l - k| \le d_n\}$, $Q^0_n = S_0 \cup \mathcal{P}^0_n$; use $\mathcal{P}_n = \mathcal{P}^0_n$ if $Q^0_n = [p]$ or $p \ge 2|Q^0_n| + 32s_0 + 1$, and $\mathcal{P}_n = [p] \setminus S_0$ otherwise. Then (H0), (G1)–(G3), (G4-dir) hold and (C4) applies (with MR1 (C-GROW)); $|Q| \le 4s_0 d_n + 34s_0 + 1 = O(\log n)$ (an envelope: it is not claimed that the start mode has unboundedly many extras). (c) *AR(1) with iid Gaussian rows = study S2*: raw rows iid $N(0, \Sigma)$, $\Sigma_{ij} = \rho^{|i-j|}$, fixed $\rho \in (-1, 1) \setminus \{0\}$, columns centred and standardised, $p \le n^A$ (any fixed $A \ge 1$; $p \gg n$ allowed); on standardised columns $Y = X\theta^* + C\varepsilon$ with fixed nonzero θ^*, $\varepsilon \sim N(0, \sigma^2 I_n)$ independent of the raw design, $\sigma^2 > 0$ fixed, and $y = Y/(\|Y\|/\sqrt{n-1})$ inside the score; separation and the piecewise relevance set as in (a); any X-measurable partition with blocks of size ≤ 16; implementation premises and MR1 (C-GROW) retained. Under the joint law of (X, ε) the three events (sandwich, NA, TL) are proved directly; the generator's S2 layouts ($p = 500, 1000$; 14 truths; $\rho = 0.5$) satisfy separation for every scale $c > 0$, so the result covers asymptotic sequences extending them — *not* the generator's

fixed $n = 200$. (b) *S5 population covariance at fixed* $p = 100$ (diagnostic layout): an illustration only — for fixed p the production run eventually enumerates all models ($n \gtrsim 2.5 \cdot 10^{27}$ for $p = 100$, before the (G4-χ) onset), so T03♯ alone gives the conclusion there; Theorem GROW adds content beyond T03♯ + enumeration only for $p \gg \log n$ [W1C/C-GROW/STATEMENT.md:118-149].

Status and evidence

C-GROW: PROVED-CHECKED (council round 1: Sol 9.2, Opus 9.0, all six gates PASS); L6 (random AR(1)) by a parallel prover (Codex Sol), checked by R07 Sol xhigh VERIFIED and R06 Opus VERIFIED-WITH-AMENDMENTS [W1C/C-GROW/REPORT.md:24-30]; statement v2.5 [W1C/C-GROW/STATEMENT.md:84-117]. Inputs: M5(0) PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5)); T03♯ as above; [OP] PROVED (by this seat, unchecked) inside GEN. Premise list keeps e4 verbatim, but the proof uses only "the emitted LS0 centre is some maximiser of γ_0". Finite n: the asymptotic events fail at tested n; S5 needs $\log n \gtrsim 96$.

5.6 Reported HPM under misspecification (C-MIS Corollaries H, D, D′)

Corollary 5.5 (H)**.** *At every n, for any candidate design and data law on valid inputs and any feasible T (deterministic or design-measurable), with* $D_n(T) := \mathbb{P}(\bar{\mathcal{O}} \cap \{T \notin \bar{S}^+\})$*:* $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq T\}) \leq 2\mathbb{E}[1 - \pi_{\mathcal{C}}(T \mid y)] + D_n(T)$*.*

Corollary 5.6 (D, D′)**.** *Under GR with a fixed candidate design, method v3.1 + LS0 with GEN's premises, and GEN's (H0)–(H2) (or (H0), (H2) with* $S^* = S_0^{\mathcal{C}}$*) for the candidate problem with* $\sigma^2 := \sigma_{\mathcal{C}}^2$ *everywhere in GEN and* $\mu := \mu^{\mathcal{C}}$*, on the actual candidate-only partition:* $D_n^{\mathcal{C}} \to 0$, $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0^{\mathcal{C}}\}) \to 0$, $\liminf \mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0^{\mathcal{C}}\}) = 1 - \limsup \mathbb{P}(U_n)$*; random candidate designs via GEN (C4).* ***D′*** *(instance on GEN's W family): the omitted variable is a noisy copy of a candidate* j_2 *that is null in the full model, and the reported HPM contains the proxy* j_2 *in place of the omitted k.*

Status: C-MIS PROVED-CHECKED; Cor. D inherits GEN's premises (LS0 as specified, e4, GUARD-W/OP-LS0, BM-∞); no delivery claim [W1C/C-MIS/STATEMENT.md:109-120].

5.7 The Song–Liang-style general-design result and study coverage

Reading for the paper. GEN and GROW together are the reported-model analog of Song–Liang's MAP consistency on general designs: Song–Liang's identifiability Condition 3 and screening Condition 5 are replaced by (i) Gram floors on a relevance set, (ii) T03♯'s anchored sparse eigenvalues and beta-min, and (iii) block-level visibility of the truths and negligibility of the outside variables on the method's own partition — bounded relevance sets with masking allowed (GEN, certificate (C1′)), or relevance sets of size $O(\log n)$ without masked truths at block level (GROW). Both are asymptotic and joint with output.

Scenario	design (generator `Scripts/Data_generators/gen_dev_data.R`)	covering result (asymptotic sequence, not $n = 200$)	status
S1, S3, S4, S6	groups of 5, iid rows, 14-signal layouts	GEN (C5)(d), favourable case via (C4), Lemma SEL (amended P-SEL.1–3)	GEN PROVED-CHECKED; SEL VERIFIED-WITH-AMENDMENTS
S2	AR(1), $\rho = 0.5$, iid rows	GROW (C5)(c), $p \leq n^A$ incl. $p \gg n$, under MR1 (C-GROW); delivery: Prop. 6.4	C-GROW PROVED-CHECKED

Scenario	design (generator `Scripts/Data_generators/gen_dev_data.R`)	covering result (asymptotic sequence, not $n = 200$)	status
S5	compound symmetry (within 0.7, between 0.3)	**frontier**: only GROW (C5)(b), fixed $p = 100$ population-covariance illustration; random-design S5 and factor designs with growing p not proved	—

Sources: [W1/W1-GEN/STATEMENT.md:135-143]; [W1C/C-GROW/STATEMENT.md:150-156]; [W1C/C-GROW/REPORT.md:12-15].

6 Delivery

All results of §5 are joint with output: they leave $\mathbb{P}(U_n)$ free. C-DLV asked whether $\mathbb{P}(U_n) \to 0$ for v3.1 + LS0 under a declared polynomial resource allocation. **For the shipped calibration this target remains open**; what is proved is an exact reduction to one calibration event, and — with a total tier-1 calibration fallback — $\mathbb{P}(\text{no output}) = 0$ (Proposition 6.3, extended to the GROW class by Proposition 6.4). The fallback is implemented on the C-ENG branch `w1c/ls0-merge` (default on, switch `v3.grid_fallback`; not merged to main), and none of the 2,400 study fits had status `schedule_limit` (§7).

Setting. Method $\mathfrak{M}$ = v3.1 + LS0 as specified, **with LS0 move cap** ℓ_n (C-DLV's CONFIG; for the C-GROW cap a_n see Proposition 6.3 and §7.1), $R' = 4k$ declared (k fixed); premises EX-A (also as a unit-cost model for exact reals, including exp and log), EX-S, SR, OP-LS0 (cap ℓ_n), e4, GUARD-W, IDEAL-W, MACH, CONFIG, TDP-IMPL; valid declared seed; **CAL is not assumed**. Classes: LH (exact Gram), LH-iid, the W1-GEN class; fixed s_0, $p \le n^A$. $\mathcal{U}_n$ = no output; $\mathcal{SL}_n$ = status `schedule_limit` [W1C/C-DLV/STATEMENT.md:10-46]; [W1C/C-DLV/lemmas/L4_delivery.md:6-9].

Allocation. In Theorem 6.1 and Corollary 6.2 the declared allocation satisfies C-DLV Corollary R: $\text{mem}_n \ge \max\{\beta\bar{M}(n, p_{\text{raw}}) + \beta_0, P_0, P_1\}$ and $\text{time}_n \ge \max\{2[\tau\bar{W}(n, p_{\text{raw}}) + \tau_0], \bar{P}_t\}$, with `allocation$mem_gb` $\ge \text{mem}_n/2^{30}$ and `allocation$hours` $\ge \text{time}_n/3600$; here $\bar{W}, \bar{M}$ are Lemma R's work and memory envelopes, $P_0, P_1, \bar{P}_t$ its preflight bounds, $\tau, \tau_0, \beta, \beta_0$ MACH's constants [W1C/C-DLV/lemmas/L1_resources.md:129-143]. These statements use LS0 cap ℓ_n; for the GROW cap use A1's amended envelopes as specified in Proposition 6.3. (A merely "valid" allocation does not suffice: a tiny memory budget stops the fit at preflight with `resource_limit`.)

6.1 The exact reduction

Theorem 6.1 (D; C-DLV). *Under the setting and allocation above, for every $n \ge 3$: (i) $\mathcal{U}_n \subseteq \mathcal{SL}_n \cup \mathcal{N}_n$ with $\mathbb{P}(\mathcal{N}_n) = 0$, and $\mathcal{SL}_n \subseteq \mathcal{U}_n$; (ii) $\mathcal{SL}_n = \{$calibration escalates to tier 1, and some tier-1 grid construction fails$\}$; (iii) a construction fails iff (α) its interval count exceeds the tier cap T_e (pilot stages plus appended bisection points), or (β) non_progress (requires a d-range on the candidate set above $2\sqrt{c_0}\,2^{60} \approx 7.5 \cdot 10^{17}$, $c_0 = \log(10/9)$), or (γ) failed validation at a point snapped to 1. Hence $\mathbb{P}(\mathcal{U}_n) \to 0 \iff \mathbb{P}(\mathcal{SL}_n) \to 0$.*

Supporting results: **(P1)** inventory of every exit of `v3_fit` without estimates (AUDITED by code reading); **(P2) Lemma R**: operations, words, preflight bytes and preflight seconds are bounded by explicit polynomials in (n, p_{raw}) for every valid input, seed and uniform realisation, so under MACH and the allocation above no resource path fires; **(P3) Lemma S**: on the classes the input/degenerate statuses have probability 0 and under EX-A no scorer call fails; **(P5)** grid lemmas (exact identity $\rho_V(u, v) = q_u(M)q_{2v-u}(M)/q_v(M)^2$; pilot schedules have $\le T_e$ strictly increasing stages ending at 1; dominance or two-model sufficient conditions) [W1C/C-DLV/STATEMENT.md:52-67].

(O) The open event, exact form. $\mathbb{P}(\mathcal{SL}_n) \to 0$ on the classes, i.e. $\mathbb{P}(E_n \cap F_n^1) \to 0$, E_n = some tier-0 trigger fires, F_n^1 = some tier-1 construction fails through (α)–(γ). Sufficient routes: (O-a) $\mathbb{P}(E_n) \to$

0; (O-b) control of (α)–(γ) for the realised tier-1 candidate sets and pilot schedule jointly with the trigger. The pilot stage count is the CJ06 / Conjecture P question; neither a pilot bound alone nor endpoint posterior concentration suffices. No design sequence is known to satisfy the (P5) sufficient conditions; in all 134 production fits run by C-DLV no tier-0 trigger fired, so tier 1 was never reached [W1C/C-DLV/STATEMENT.md:68-84].

Corollary 6.2 (D′: joint consistency with the delivery term)**.** *On LH, LH-iid and the GEN class, under the respective hypotheses of W1-GEN (C1)+(C2) with T03♯, and W1-PXN (T), (P), with LS0 cap ℓ_n and the allocation above: $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) \geq 1 - o(1) - \mathbb{P}(\mathcal{SL}_n)$, pathwise from $\mathcal{O} \cap \{\hat{S}^+ \neq S_0\} \subseteq \{\pi(S_0 \mid y) \leq \frac{1}{2}\} \cup \mathcal{K}_n^c$, $\mathcal{K}_n$ = the prescribed deterministic LS0 scores S_0. CAL is not used; the LH exact-Gram instance carries the input status SAME-FAMILY-ONLY.*

Status and evidence

C-DLV: PARTIAL — the ticket target (T) "$\mathbb{P}$(no output) → 0 … for v3.1 + LS0 under a declared polynomial allocation" remains open for the shipped calibration. The reduction package: council round 2 passed (Codex 9.0, Opus 8.9, all six gates PASS) [W1C/C-DLV/REPORT.md:22-27]. Per clause: P1 AUDITED (code reading); P2 PROVED-CHECKED (council round 2); P3 S.1, S.2 PROVED-CHECKED, S.3 AUDITED (code reading); P4 PROVED-CHECKED (relative to the P1/P3 audits); P5 PROVED-CHECKED; C2 PROVED-CHECKED; O OPEN [W1C/C-DLV/REPORT.md:61-70]. Proofs `W1C/C-DLV/lemmas/L1_resources.md–L5_insertions.md`; Theorem D [W1C/C-DLV/lemmas/L4_delivery.md:13-22].

6.2 Proposition F: the total tier-1 fallback

C-DLV found (as a finding on a shipped component, not made by the ticket) that `schedule_limit` is the only no-output path not excluded by the declared allocation and the idealisation premises, and that freezing a valid fallback grid removes it [W1C/C-DLV/lemmas/L4_delivery.md:79-93]. The HQ red-team audit A1 found HOLDS-WITH-AMENDMENTS and no fatal finding; its one MAJOR finding is that the combination with C-GROW's LS0 cap needs amended resource envelopes and allocation (the old time shape is not justified after the change: at $p \propto n$ the old core term is of order $n^5 \log^3 n$, while the sufficient extra LS0 term is of order n^6) [W1C/HQ_AUDIT/A1_propF_redteam.md:7-19]. The statement below is A1's "Exact corrected statement" [W1C/HQ_AUDIT/A1_propF_redteam.md:60-75].

Proposition 6.3 (F: total tier-1 fallback, with the C-GROW cap; C-DLV + A1)**.** *Consider the specified v3.1 method with deterministic LS0 from the exact e4 start mode, complete scans, emitted centre and scored-set HPM retention. Let $\ell_n = \lceil \log_2(n + p + 1) \rceil$, $K = \min(p, \lfloor n/2 \rfloor)$, and set LS0's move cap to $a_n = \ell_n + |\tilde{S}_0|$; all other local-search caps remain ℓ_n. Keep fixed declared $R' = 4k$, fixed worker count, and the other CONFIG restrictions of C-DLV. Assume EX-A with its unit-cost interpretation, IDEAL-W, EX-S, SR, e4, GUARD-W, TDP-IMPL and MACH as stated in C-DLV, and OP-LS0 with the cap replaced by a_n, with valid declared seed/allocation and model G on the stated classes.*

Fallback. *At the first failed tier-1 grid construction, terminate that tier and select its last successful construction if one exists, otherwise its completed pilot schedule. Preserve the tier-1 budgets and inherited scored state, freeze the selected complete grid through the normal frozen-record construction before final randomness, and continue with the prescribed final run and LS0. Retain the failure/fallback metadata; make no further calibration attempts and do not promote a partial failed grid to a fallback. Leave other branches unchanged.*

Allocation. *With $q(p) = p + p^2/4$, $W_G := \bar{W} + C\,K\,q(p)\,[n^3 + p(n+2)]$ and $M_G := \bar{M} + C\,K(K+1)$ ($\bar{W}, \bar{M}$ the work and memory envelopes of C-DLV Lemma R, C a large enough constant for the fixed implementation and concurrency (fixed worker count), raw dimension in the envelopes), choose*

$$\mathrm{mem}_n \geq \max\{\beta M_G(n, p_{\mathrm{raw}}) + \beta_0, P_0, P_1\}, \quad \mathrm{time}_n \geq \max\{2[\tau W_G(n, p_{\mathrm{raw}}) + \tau_0], \bar{P}_t\},$$

`allocation$mem_gb` $\geq \mathrm{mem}_n/2^{30}$, `allocation$hours` $\geq \mathrm{time}_n/3600$, where $P_0, P_1, \bar{P}_t$ are Corollary R's preflight bounds and certificate budgets remain included. Addition for the C-ENG head code (council

round 3, Opus finding 1): *for the implementation at `493579a`, whose preflight also charges LS0's worst case (Remark (5)), the allocation must in addition satisfy* $\text{time}_n \geq \bar{P}_{\text{LS0}} := (\ell_n + K + 1)q(p)(\tau U_K + \tau_0)/100$ *(U_K the operation count of the 200 timed size-K scores) and* $\text{mem}_n \geq P_e$ *+ the `ls0_archive` bytes ($e = 0, 1$).*

Then for every admissible $n \geq 3$, $\mathbb{P}(\text{no output}) = 0$. On LH exact Gram, LH-iid and the W1-GEN class, under the respective statistical hypotheses and inherited dependency qualifications used in Corollary 6.2 (for random-design members of the GEN class, model G is read as: X^r has nonconstant columns almost surely and $\varepsilon \sim N_n(0, \sigma^2 I_n)$ is independent of X^r; Lemma S.1 verifies the first for fixed standardised designs and nonsingular Gaussian rows), $\mathbb{P}(\hat{S}^+ = S_0) \to 1$ under the original unconditioned data law. CAL, restricted-ESS success, and any bound on the probability of triggering the fallback are unnecessary. With the original LS0 cap $a_n = \ell_n$, the original Lemma R envelopes suffice.

Status and evidence

Proposition F with cap ℓ_n (C-DLV form): PROVED-CHECKED for the patched calibration law (R04; council round 2) [W1C/C-DLV/REPORT.md:70]. The corrected form above (total fallback, cap a_n, envelopes W_G, M_G) is HQ audit A1's (fresh Codex Sol xhigh red team: HOLDS-WITH-AMENDMENTS, no FATAL finding; checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6)). A1 also independently confirms that the larger-cap LS0 path contains the smaller-cap scored set [W1C/HQ_AUDIT/A1_propF_redteam.md:56].

6.3 Extension to the GROW class (A2, checked by A3)

Proposition 6.4 (F on the GROW class; A2 with A3's amendments)**.** *Under exactly the method, fallback, premises and allocation of Proposition 6.3 (cap $a_n = \ell_n + |\tilde{S}_0|$, i.e. MR1 (C-GROW)), add to its classes: (i) fixed designs under GROW's G, (H0), (G1)–(G4), including exact-Gram AR(1) under all the hypotheses of (C5)(a); (ii) iid-row AR(1) under all the hypotheses of (C5)(c), including independent Gaussian noise and the separation condition (the iid instance uses its joint-law event theorem, not fixed-design NEG or (G4-dir)) [W1C/HQ_AUDIT/A2_propF_GROW.md:25]. Class (i) contains the fixed-p illustration (C5)(b); for every fixed-p sequence the conclusion is eventually delivered by enumeration plus T03♯. Random-row S5 (study S5) and the (C6) frontier are not covered. Then, for every admissible $n \geq 3$, $\mathbb{P}(\text{no output}) = 0$, and along each such sequence $\mathbb{P}(\hat{S}^+ = S_0) \to 1$, under the original unconditioned noise law (fixed designs) or the original joint design/noise law (iid rows), together with the algorithm's randomness. CAL, candidate restricted-ESS success and a bound on the fallback probability are unnecessary.*

Proof outline (A2, with A3's amendments). (1) *Lemma S extends*: GROW's fixed designs have nonconstant raw columns; for iid AR(1) each raw column is iid $N(0,1)$, so nonconstant a.s.; no column is dropped, so the retained p equals p_{raw} a.s. and the allocation depends only on (n, p_{raw}); a constant centred response is a null event (for iid rows, $C\varepsilon$ is nondegenerate on $\mathbf{1}^\perp$ given X); the scorer never fails since $Q_S \geq y^\top y/(n+1) > 0$, with no rank or relevance-size condition, so $p \gg n$ causes no numerical obstruction; S.3 keeps its AUDITED status with "successful construction" read as "successful construction or complete fallback pilot". (2) *Delivery*: MR1 (C-GROW) permits at most $\ell_n + K + 1$ LS0 scans, i.e. at most $Kq(p)$ extra score requests; in C-DLV `W1C/C-DLV/lemmas/L4_delivery.md` line 38 and `W1C/C-DLV/lemmas/L1_resources.md` lines 25 and 43, replace ℓ for LS0 by $a_n \leq \ell_n + K$, so the LS0 count becomes $\leq 1 + (\ell_n + K + 1)q(p)$ and $\bar{\Sigma}$ becomes $\bar{\Sigma} + Kq(p)$; Corollary R then applies with W_G, M_G, and with the total fallback $\mathcal{SL}_n = \emptyset$, so Theorem D's walk leaves only null exits. (3) *Consistency*: with $\mathcal{K}_n$ = "deterministic LS0 with cap a_n scores S_0", on $\text{Sand}_n \cap \text{NA}(\mathcal{L}) \cap \text{TL}(\mathcal{L})$ the start mode lies in $\mathcal{L}$ and LS0 deletes one extra per move, so $\mathbb{P}(\mathcal{K}_n^c) \leq \mathbb{P}(\text{Sand}_n^c) + \mathbb{P}(\text{NA}^c) + \mathbb{P}(\text{TL}^c) \to 0$ (GROW's descent on Sand ∩ NA ∩ TL and the cap check [W1C/C-GROW/lemmas/L4_assembly.md:26-28] and the event limits [W1C/C-GROW/lemmas/L4_assembly.md:28-29]); this is *not* inferred from $D_n \to 0$, which [OP] bounds in the opposite direction. T03♯ under (H0) (fixed design), or L6.2's design events with the diagonal argument (iid rows), gives $\mathbb{E}[1 - \pi(S_0)] \to 0$. D''s pathwise claim and $\mathbb{P}(\mathcal{O}) = 1$ give $\mathbb{P}(\hat{S}^+ \neq S_0) \leq 2\mathbb{E}[1 - \pi(S_0)] + \mathbb{P}(\mathcal{K}_n^c) \to 0$ [W1C/HQ_AUDIT/A2_propF_GROW.md:37-67].

Status and evidence

PROVED (A2), checked by A3; checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6). A2: Codex prover, verdict EXTENDS [W1C/HQ_AUDIT/A2_propF_GROW.md:1-5]; A3: cross-family check (Claude Opus 5.5) VERIFIED-WITH-AMENDMENTS, five MINOR amendments (score-count replacement, S5/fixed-p scope, p_{raw} wording, status enumeration, citation range), all applied above [W1C/HQ_AUDIT/A3_check_A2.md:5]. Inherited qualifications (A3 finding 4): A1's corrected statement was not councilled at the time of A3 (since checked in full by the final council, round 3); GROW's L4 and L6 are PROVED (by this seat, unchecked) with checks R04 and R06/R07; [OP] is PROVED (by this seat, unchecked); e4 rests on N3, which is OPEN (only "some maximiser" is used here, which the coded routine and TDP satisfy); S.3 is a code-reading audit; the fallback is a change to a shipped component, implemented on the unmerged C-ENG branch [W1C/HQ_AUDIT/A3_check_A2.md:31-39].

Remark 6.5 (what Propositions F and F-GROW do and do not say)**.** (1) The fallback changes the calibration law only on the event where the shipped code returns no estimates; final-run validity holds conditionally on any frozen grid (calibration contract). The fallback grid keeps the contract's T_e cap but need not have the candidate grid's restricted-ESS property. (2) Implementation precision (A1): the shipped code sets `out$grid=NULL` on *any* grid failure, so changing only the status line would fail validation; the patch must carry a complete grid through the tier return and the normal freeze, and must not use a failed construction's partial prefix [W1C/HQ_AUDIT/A1_propF_redteam.md:43]. (3) It is a theorem about an abstract unit-cost, unbounded-word machine with MACH's allocation-success and environment clauses; it is *not* a reliability statement about the shipped binary or the default 8 GiB / 24 h allocation. The exclusion of every other no-output path is relative to the P1 exit inventory and Lemma S.3, both AUDITED (code reading) on the serial shipped call graph; the LS0, worker and fallback paths are covered by premises, not by that audit [W1C/HQ_AUDIT/A1_propF_redteam.md:33]. (4) The classes are LH, LH-iid, the GEN class (Proposition 6.3) and the GROW class including iid AR(1) (Proposition 6.4); not the C-MIS candidate-rerun setting, random-row S5 or factor designs with growing p. (5) Implementation: C-ENG implements exactly this fallback (last successful tier-1 construction, else the completed tier-1 pilot; frozen through the ordinary `v3_frozen`; flag `grid_fallback` only when it fires), reviewed APPROVE, on the unmerged branch [BR/Theory_work/Wave1C/C-ENG/REPORT.md:54]; at the reviewed head `8a27652` the preflight formula did not include the MR1 (C-GROW) LS0 term [BR/Theory_work/Wave1C/C-ENG/REPORT.md:62-64]; at the current branch head `493579a` (commits `ef33f91`, `98fac03`, `72785fe`; reviews R07 REJECT then fixed, R08 APPROVE-WITH-FIXES, R09 APPROVE; no production change after `72785fe`) the SMC preflight charges the LS0 worst case when the MR1 (C-GROW) cap is in force: an archive memory component `ls0_archive` and a time estimate $2(\ell_n + K + 1)q(p)\hat{s}_K$ ($\hat{s}_K$ the measured seconds per score of a size-K model, 200 timed scores) checked against `allocation$hours`; with `v3.ls0_cap = "ell_n"` or `ls0 = FALSE` the shipped preflight is unchanged [BRH/Theory_work/Wave1C/C-ENG/REPORT.md:63-81]; [BRH/Scripts/SMC_BVS/core/v3_fit.R:43-65]. This is a new preflight exit outside C-DLV's P1 inventory; an allocation that also satisfies the additional preflight terms in Proposition 6.3's allocation is not refused by it. The writer has not audited the branch code against the proposition beyond the C-ENG report and the LS0 lines cited in §7.1.

7 Implementation status and method requirements

LS0 and the e4 tie fallback were not in the production core; the Wave-1C protocol allowed proofs to fix parameters of these not-yet-shipped components as method requirements, and forbade changes to shipped components (reported as findings instead) [W1C/PROTOCOL.md:40-47]. Two different requirements carry the label "MR1": **MR1 (C-PACK)** concerns the LS0 *centre* (the coded start mode), **MR1 (C-GROW)** the LS0 *move cap*.

C-ENG (implementation). Ticket C-ENG implemented all forwarded requirements on branch `w1c/ls0-merge` (pushed to origin; reviewed head `8a27652`, current head `493579a`, which adds only the

LS0 worst-case preflight charge; base `f16ce15`; **not merged to main**); result "DONE (implemented, tested, all code reviews APPROVE)" [BR/Theory_work/Wave1C/C-ENG/REPORT.md:3-8]. Facts used below, all from that report (path `origin/w1c/ls0-merge:Theory_work/Wave1C/C-ENG/REPORT.md`):

- Everything is behind switches; the default is the new behaviour; `v3_fit(…, ls0 = FALSE)` reproduces the v3 fit entirely (checked 48/48 on the final code); per-item switches `v3.ls0_cap = "ell_n"`, `v3.grid_fallback = FALSE`, `v3.ls0_tie_dp = FALSE` restore the old rule of each item [BR/Theory_work/Wave1C/C-ENG/REPORT.md:12-16].
- LS0 runs after the final local search and its guard check, only on a completed SMC run, from `start_mode(st, 1e6, tie_dp)` with emitted centre; it does not exist on the exact branch; the e4 truncation fallback is TDP-E (returns S^* under exact arithmetic) [BR/Theory_work/Wave1C/C-ENG/REPORT.md:38]; [BR/Theory_work/Wave1C/C-ENG/REPORT.md:73-78].
- MR1 (C-GROW): LS0 cap $= \ell_n + |\tilde{S}_0|$, applied only to LS0; the final local search keeps ℓ_n [BR/Theory_work/Wave1C/C-ENG/REPORT.md:52]. At `8a27652`, the shipped preflight formula was *not* updated for the larger cap ("handled on the theory side") [BR/Theory_work/Wave1C/C-ENG/REPORT.md:62-64]; at the current branch head `493579a` (commits `ef33f91`, `98fac03`, `72785fe`; reviews R07 REJECT then fixed, R08 APPROVE-WITH-FIXES, R09 APPROVE; no production change after `72785fe`) the SMC preflight charges the LS0 worst case when the MR1 (C-GROW) cap is in force: an archive memory component `ls0_archive` and a time estimate $2(\ell_n + K + 1)q(p)\hat{s}_K$ ($\hat{s}_K$ the measured seconds per score of a size-K model, 200 timed scores) checked against `allocation$hours`; with `v3.ls0_cap = "ell_n"` or `ls0 = FALSE` the shipped preflight is unchanged [BRH/Theory_work/Wave1C/C-ENG/REPORT.md:63-81]; [BRH/Scripts/SMC_BVS/core/v3_fit.R:43-65] (see Remark (5) of §6).
- C-DLV B tier-1 fallback with A1's precision: the last successful tier-1 construction, else the completed tier-1 pilot schedule (never the failed or partial grid), validated and frozen through the ordinary `v3_frozen`; flag `grid_fallback` set only when the fallback fires [BR/Theory_work/Wave1C/C-ENG/REPORT.md:54].
- Study outputs: the 2,400 study fits contain 0 with status `schedule_limit`, so the published outputs are unaffected by the fallback [BR/Theory_work/Wave1C/C-ENG/REPORT.md:56-60].
- Code reviews R03, R04, R06 APPROVE; R05 APPROVE-WITH-FIXES, fixes checked by R06 [BR/Theory_work/Wave1C/C-ENG/REPORT.md:80-92]. Out of scope of C-ENG: the frozen study core (unchanged), the certificate card, stacking / the c_4 search, the concurrency layer, Owls-scale runs [BR/Theory_work/Wave1C/C-ENG/REPORT.md:190-201].

Requirement	content and reason	source	implementation (C-ENG branch `8a27652`)
LS0 (OP-LS0)	one §6.10 local search from the start mode after the final search (and after the c_4 search when $R' > 4$), centre emitted, no uniforms, provenance `ls0`, GUARD-W semantics	SoP §6.10 / METHOD #8	implemented, default on; switch `ls0`
e4 → e4′; MR1–MR3 (C-PACK)	LS0 centre = coded `start_mode(st,cap).S` (or a routine returning the unique maximiser when unique) as a data function on every branch; **no consistency result needs the lexicographic tie rule or the truncation fallback**; GROW needs the emitted centre to be *some* maximiser of γ_0	C-PACK [W1C/C-PACK/METHOD_REQUIREMENTS.md:6-16]	implemented (MR2: cap $<$ 2 rejected); size-then-lex rule kept for reproducibility
TDP-IMPL	the e4 tie fallback is W1-N3's reference algorithm TDP (or returns the same maximiser within its bounds), $O(p\sum_b(2^{\lvert C_b\rvert} + (K+1)^2))$ operations; needed only for the delivery resource bound	C-DLV A.2 [W1C/C-DLV/METHOD_REQUIREMENTS.md:8-15]	TDP-E, same maximiser under EX-A; switch `ls0_tie_dp`

Requirement	content and reason	source	implementation (C-ENG branch `8a27652`)
MR1 (C-GROW): LS0 cap	$\ell^{\mathrm{LS0}} = \ell_n + \lvert\tilde{S}_0\rvert$ (final search keeps ℓ_n); for GROW alone any cap $\geq \lvert\tilde{S}_0\rvert$ is accepted; **all results are established simultaneously for caps** $\geq \max(\ell_n, \lvert\tilde{S}_0\rvert)$ **(and** $\leq \ell_n + K$ **for the delivery envelopes); none is claimed for smaller caps** (Lemma 7.1 needs $a \geq \ell_n$); a cap $o(\lvert\tilde{S}_0\rvert)$ cannot carry the LS0 descent guarantee when $\lvert\tilde{S}_0 \setminus S_0\rvert \to \infty$	[W1C/C-GROW/METHOD_REQUIREMENTS.md:1-25]	implemented, default on; switch `ls0_cap`
Calibration fallback (shipped component)	freeze the last successful tier-1 construction or the completed tier-1 pilot schedule instead of returning `schedule_limit` (Prop. 6.3, A1's total-fallback form)	C-DLV B [W1C/C-DLV/METHOD_REQUIREMENTS.md:17-25]	implemented by HQ decision, default on; switch `grid_fallback`

Nothing in this chapter asserts that the branch code satisfies EX-A, GUARD-W or the other named premises; C-ENG's evidence is tests and code review. The premises (L1)–(L3) of Lemma 7.1 are checked against the branch code below.

7.1 Integration lemma: the larger LS0 cap

PXN, GEN, C-MIS Cor. D and C-DLV (Theorem D, Cor. D′, Prop. F in its C-DLV form) were proved with LS0 cap ℓ_n; C-GROW and the corrected Proposition F use $a_n = \ell_n + \lvert\tilde{S}_0\rvert \geq \ell_n$. The lemma below shows which conclusions transfer.

Setting. For a finite-integer-valued deterministic measurable function $a = a(X, y) \geq \ell_n$ of the admissible data (the coded cap is an `int`), let $\mathfrak{M}(a)$ be the method that differs from the specified v3.1 + LS0 only in LS0's move cap, which is a. Structural facts used, from the specification and checked on the C-ENG branch:

(L1) LS0 is the last search: it runs after the final local search and after the c_4 search ($R' > 4$); nothing computed before it depends on its cap, and its outputs enter the HPM running best, the stores, flags, archives and the post-hoc certificate card [SoP/02_method.tex:490-494]. On the branch: the four final islands, then the final local search and its guard check, then (only if the run is still `completed`) start mode, LS0 and its guard check; the function then assembles its output [BR/Scripts/SMC_BVS/core/cpp/v3_core.cpp:443-481].

(L2) LS0 reads no uniforms and alters no particle; its centre $\tilde{S}_0$ is the e4′ data function computed before LS0 and does not depend on the cap. On the branch the cap is computed from the returned start mode, $\ell_n + \lvert\tilde{S}_0\rvert$, after `start_mode` has returned [BR/Scripts/SMC_BVS/core/cpp/v3_core.cpp:469-476]. The branch also checks the cap range (`stop("LS0 move cap out of range")` if $\ell_n + \lvert\tilde{S}_0\rvert$ exceeds `INT_MAX`); this cap-dependent exit is excluded by IDEAL-W and does not exist in the specified method, to which the lemma refers [BR/Scripts/SMC_BVS/core/cpp/v3_core.cpp:474].

(L3) LS0 executes `local_search(centre, ell, …, emit_centre = true)`: the centre is evaluated; each loop iteration completes a scan of every feasible flip and swap of the current centre C, selects the best scanned neighbour (largest F, exact ties lexicographic), and then: if its F is not strictly larger than $F(C)$, terminates with `no_improvement`, `local_certified`, returning C; otherwise, if the move count has reached `ell`, terminates with `local_limit`, returning that neighbour; otherwise moves to it. The cap enters *only* in this test [CORE/cpp/v3_engine.h:385-425]; same code in [PROD/cpp/v3_engine.h:378-418] and on the branch (only a phase-label argument added) [BR/Scripts/SMC_BVS/core/cpp/v3_engine.h:379-414].

(L4) EX-A (scores exact; F a fixed function on Ω); GUARD-W (on $\mathcal{O}$ the delivered record equals the potential record); the reported HPM is the $\succ$-maximum of the all-scored set.

Let $N(C)$ be the feasible flip/swap neighbourhood of C, and let $C_0 = \tilde{S}_0, C_1, \ldots$ be the *uncapped* greedy path (C_{k+1} = best scanned neighbour of C_k when it strictly improves F); m^* = the number of its moves. Since F strictly increases along the path and Ω is finite, $m^* < \infty$.

Lemma 7.1 (CAP: LS0-cap monotonicity)**.** *Let a, a' be finite-integer-valued measurable functions of the admissible data with $\ell_n \le a \le a'$; couple $\mathfrak{M}(a)$ and $\mathfrak{M}(a')$ through the same valid seed and ideal uniforms. The deterministic LS0 search is defined on every admissible data set, also on branches that do not execute it (exact-branch convention); its scored set is distinguished from the set actually scored by the fit. For every data set, seed and uniform realisation:*

(a) Prefix. *LS0 with cap a makes $\min(a, m^*)$ moves along $C_0, C_1, \ldots$, scans exactly $C_0, \ldots, C_{\min(a,m^*)}$, and its scored set is $\mathcal{S}_{\mathrm{LS0}}(a) = \{C_0\} \cup \bigcup_{k \le \min(a,m^*)} N(C_k)$. It terminates certified at C_{m^*} iff $m^* \le a$; otherwise with `local_limit`. Hence $\mathcal{S}_{\mathrm{LS0}}(a) \subseteq \mathcal{S}_{\mathrm{LS0}}(a')$, and if the cap-$a$ search ends certified, the cap-a' search is identical (same moves, scans, scored set, return value and flags).*

(b) Potential run. *The potential runs of $\mathfrak{M}(a)$ and $\mathfrak{M}(a')$ coincide up to LS0, $\bar{\mathcal{O}}$ does not depend on the cap, $\bar{\mathcal{S}}^+(a) \subseteq \bar{\mathcal{S}}^+(a')$, $\{LS0(a') \text{ misses } S_0\} \subseteq \{LS0(a) \text{ misses } S_0\}$, and $D_n(a') \le D_n(a)$.*

(c) Reported HPM. *For $\mathfrak{M}(a)$ with any $a \ge \ell_n$: $\mathcal{O}(a) \cap \{\hat{S}^+(a) \ne S_0\} \subseteq \{\pi(S_0) \le \frac{1}{2}\} \cup \big(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+(\ell_n)\}\big)$, hence*

$$\mathbb{P}\big(\mathcal{O}(a) \cap \{\hat{S}^+(a) \ne S_0\}\big) \le 2\,\mathbb{E}_0[1-\pi(S_0)] + D_n(\ell_n) \le 2\,\mathbb{E}_0[1-\pi(S_0)] + \mathbb{P}(LS0(\ell_n) \text{ misses } S_0).$$

Proof. (a) Induction on loop iterations using (L3). Before the $(k+1)$-st scan both runs are at the same centre C_k with the same move count k (true for $k = 0$: the same emitted centre, (L2)). The scan, the selected neighbour and the strict-improvement test depend only on C_k and F ((L4)), not on the cap. If there is no strict improvement ($k = m^*$), both terminate certified at C_k. Otherwise the cap-a run terminates iff $k \ge a$, i.e. $k = a$ (the count never exceeds the cap); while $k < a$ both move to C_{k+1}. So the cap-a run scans $C_0, \ldots, C_{\min(a,m^*)}$ and the scored set is as stated; the cap-a' run scans a superset of centres. If $m^* \le a \le a'$ both stop at the same iteration with the same outputs. (b) By (L1)–(L2) everything before LS0 in the potential run uses the same uniforms and the same deterministic computations for both caps, and LS0 changes neither particles nor the calibration; the potential status (`exact`, `completed` or `schedule_limit`) is fixed before LS0, and a capped LS0 returns output, so $\bar{\mathcal{O}}$ is the same event. On a completed SMC potential run, $\bar{\mathcal{S}}^+(a) = \bar{\mathcal{S}} \cup \mathcal{S}_{\mathrm{LS0}}(a)$ with the same pre-LS0 set $\bar{\mathcal{S}}$, so (a) gives the inclusions. On the exact branch every feasible model is already scored and the sets coincide for both caps. On a `schedule_limit` branch neither fit executes LS0, the actual scored sets coincide, and the branch is outside $\bar{\mathcal{O}}$. Thus the inclusions hold on every branch, and $D_n(a') = \mathbb{P}(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+(a')\}) \le D_n(a)$. (c) Actual output implies potential output ($\mathcal{O}(a) \subseteq \bar{\mathcal{O}}$, the guard only stops), and on $\mathcal{O}(a)$ the delivered all-scored set is $\bar{\mathcal{S}}^+(a) \supseteq \bar{\mathcal{S}}^+(\ell_n)$ (GUARD-W, (b)). If S_0 lies in it and $\pi(S_0) > \frac{1}{2}$ then $F(S_0) > F(S)$ for every $S \ne S_0$, so the $\succ$-maximum is S_0. This gives the inclusion; Markov's inequality gives $\mathbb{P}(\pi(S_0) \le \frac{1}{2}) \le 2\mathbb{E}_0[1-\pi(S_0)]$, and $D_n(\ell_n) \le \mathbb{P}(\mathrm{LS0}(\ell_n) \text{ misses } S_0)$ is the operational closure [OP] of §5.1. □

Flagged for review — Lemma CAP

Status: PROVED (by this seat); checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6); cross-family check by the fidelity reviewer F1 (Codex gpt-6.1-sol, high): VERIFIED-WITH-AMENDMENTS (finite integer caps; explicit bookkeeping for the exact and `schedule_limit` branches; GEN transfer wording), all applied [W1C/FINAL/reviews/F1_fidelity_sol.md:1-12]. Council round 1: both seats found (a)–(c) correct (Sol also ran 115,500 exhaustive control-flow comparisons, all passing) [W1C/FINAL/reviews/C1_council_sol.md:22-30]; [W1C/FINAL/reviews/C1_council_opus.md:21]; rounds 1 and 2 did not pass overall (round 2: Sol 7.9, Opus 8.8, commit `10b2090`; disposition `W1C/FINAL/reviews/C2_disposition.md`); the final council, round 3 (Sol 9.3, Opus 8.6, all six gates PASS, commit `25492e4`), passed the chapter, with both seats checking Lemma CAP [W1C/FINAL/reviews/C3_council_sol.md:3]. (L1)–(L3) are now read off the C-ENG branch code (citations above). HQ audit A1 inde-

pendently states the prefix property [W1C/HQ_AUDIT/A1_propF_redteam.md:56].

What survives under MR1 (C-GROW) (cap $a_n = \ell_n + |\tilde{S}_0|$). By Lemma 7.1(c), every conclusion that was obtained by bounding $\mathbb{P}(\mathrm{LS0}(\ell_n)$ misses $S_0)$ or $D_n(\ell_n)$ and $\mathbb{E}_0[1-\pi(S_0)]$ holds verbatim for $\mathfrak{M}(a_n)$, with $\mathcal{O}$, U_n those of $\mathfrak{M}(a_n)$:

- PXN (C) and (T)'s first clause ("LS0 scores S_0"); (T)'s second clause ("returns S_0 with `local_certified`") too, because on that event the cap-ℓ_n search ends certified and (a) makes the cap-a_n search identical; PXN (K)'s *on-event* clause ($c+1$ scans on $\mathcal{T}_n$) for the same reason.
- GEN (C2) consequences, (C3), (C3-fav), (C4) and the instances (C5)(a)–(d): these retain their hypotheses and onsets; whenever $\ell_n \geq 2^{\bar{\nu}} - 2$ (part of GEN's onset), also $a_n \geq 2^{\bar{\nu}} - 2$, so the discovery conclusion transfers; more directly, the capped search scores S_0 when $a_n \geq \tau(C) - 1$ and has certified endpoint S_0 when $a_n \geq \tau(C)$.
- C-MIS Corollaries D and D′ (through GEN); Corollary H holds for every cap as stated. M5(b1-LH) (LS-score: LS0 scores S_0 within $\max\{\Phi(S^\circ) - 1, 0\}$ moves) — with its own statuses unchanged.
- The *pathwise* part of C-DLV Corollary D′, $\mathbb{P}(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2\mathbb{E}[1-\pi(S_0)] + \mathbb{P}(\mathcal{K}_n(\ell_n)^c)$ (since $\mathcal{K}_n(a_n) \supseteq \mathcal{K}_n(\ell_n)$), and the consistency clause of Proposition 6.3. D′'s *delivery* form ($\geq 1 - o(1) - \mathbb{P}(\mathcal{SL}_n)$) and Theorem 6.1(i) hold for $\mathfrak{M}(a_n)$ only under A1's amended allocation (W_G, M_G). SL2's discovery input ($\mathbb{P}(\mathcal{O} \cap \{S_0 \notin \mathcal{S}^+\}) \to 0$); M4(c) retention is a property of the update rule $\mathrm{top}_c(\mathcal{T}_{\mathrm{old}} \cup \mathcal{S}_{\mathrm{added}})$, not of the cap.

These transfers require $a \geq \ell_n$; a cap below ℓ_n (e.g. the cap $|\tilde{S}_0|$ when $|\tilde{S}_0| < \ell_n$) would keep GROW but could lose the PXN/GEN discovery guarantees (GEN needs $a \geq 2^{\bar{\nu}} - 2$, PXN (T) needs $a \geq c$).

What does not transfer verbatim.

- *Worst-case costs.* PXN (K)'s first clause becomes $\leq 1 + (a_n+1)\max_{k \leq K}[p + k(p-k)] \leq 1 + (\ell_n + K + 1)(p + p^2/4)$ scores — of order $(\ell_n + K)p^2$, not $O(p^2 \log(n+p))$ in general. C-DLV's Lemma R row "LS0 … cap $\ell_n = \ell$", its score count, work/memory envelopes and Corollary R's allocation must be replaced by A1's W_G, M_G and amended allocation (built into Prop. 6.3); the old declared allocation is not shown to prevent a guard stop [W1C/C-DLV/lemmas/L1_resources.md:24-25]. At `8a27652`, the branch's preflight formula still used the old LS0 count; at the current head `493579a` it charges the LS0 worst case (archive bytes and the measured-time estimate $2(\ell_n + K + 1)q(p)\hat{s}_K$), so Proposition 6.3's allocation carries the additional terms stated there (see Remark (5) of §6). M8's accounting (card TH08) counts local searches with $\ell = \lceil \log_2(n+p+1) \rceil$ [SoP/04_theory.tex:708]; its LS0 term would have to be re-charged with $a_n + 1$ scans.
- *Flags and metadata.* When the cap-ℓ_n search ends with `local_limit`, the cap-a_n search continues: `local_limit`, the LS0 move count and return value, `ls0_improved`, the `local_certified` flag of the finally reported model, the provenance counts, the score-event ledger, the store contents beyond S_0 (the update top_c can evict models, so stores are not monotone in the cap), the near-tie list and the certificate card's store-based quantities may all differ. No theorem of this chapter asserts these outside the events named above.
- *Negative statements.* GEN (C6) ("P(LS0 from $\tilde{S}_0$ misses S_0) $\to 1$" on its explicit design) is about the cap ℓ_n: its proof uses that LS0 reaches only models within symmetric-difference distance $2(\ell_n+1)$ of $\tilde{S}_0$. Monotonicity runs the other way, so (C6) is **not established for MR1 (C-GROW)**; there the reachable distance is $2(a_n+1) = 2(\ell_n + |\tilde{S}_0| + 1)$.
- *The actual no-output event.* U_n depends on the cap through runtime (a longer LS0 can meet a guard stop, which on the branch is checked after the completed LS0). Joint-with-output statements are stated for each method's own U_n; delivery requires the amended allocation (Prop. 6.3).

8 Scope and frontier

Not claimed / not proved — what the paper must not claim

1. **Selection consistency is asymptotic.** No quantitative finite-n selection-consistency guarantee is proved, in particular none for the study. (Deterministic inequalities such as SL0, M5(0) and Lemma CAP, the sampler identities N1/Corollary E, and the ideal-machine delivery statement of Proposition 6.3 keep their stated finite-n scope.) Onsets: the TH09 binding condition first holds near $n \approx 2 \cdot 10^6$ for the exhibited families (an illustration, not a universal onset) [SoP/04_theory.tex:811]; the LH-iid onset of $L'_n \delta_n \le \lambda_0/2$ is $\approx 10^{10}$ for typical constants (C-PACK council) [W1C/C-PACK/lemmas/I2_LHiid_unconditional.md:80-81]; PXN's proven margins are vacuous at $n = 200$; GEN's visibility margins at the study size are 0.16–0.94 (< 1); GROW's asymptotic events fail at tested n and S5 needs $\log n \gtrsim 96$. **The study ($n = 200$) is not covered** by any asymptotic result; there, M5(0)'s finite-n composition bracket (with the posterior-miss probability measured, not bounded) is the account.
2. **Polynomial dimension only**: $p \le n^A$ for fixed A; no uniformity in growing A; exponential p (Song–Liang Cor. 1 regime) is open (CJ18).
3. **Growing relevance sets of factor type**: S5 / common-factor designs with growing p and random-design S5 are not proved (GROW (C6)); obstacles: the sandwich needs $|Q| = O(\log n)$ and NA needs a factor-aware direction count [W1C/C-GROW/REPORT.md:86-90]. Growing s_0 is not covered anywhere.
4. **Delivery for the shipped calibration** is open: event (O) of Theorem 6.1. Delivery is proved only for the fallback calibration of Proposition 6.3, under IDEAL-W, MACH, CONFIG and the amended allocation, on LH, LH-iid and the GEN class (Proposition 6.3) and on the GROW class (Proposition 6.4, PROVED (A2), checked by A3; checked in full by the final council, round 3 PASSED (Sol 9.3, Opus 8.6)); not for the C-MIS rerun setting, random-row S5 or factor designs.
5. **Reported pooled PIPs**: no in-probability sure-screening theorem for the shipped $R' = 4$ pooled PIPs; only the islandwise bound, M1(i) under stacking ((h), (i): PROVED (by this seat, unchecked)) and the store-restricted SL2 (§4.2).
6. **Misspecification**: exact-model concentration is proved under GR (fixed τ^2, fixed projected sparsity, T03♯ hypotheses on the projected truth), and for fixed candidate designs under EP with the hypotheses of Theorem 3.6, including its size-tail condition or the stated bounded-dimension/TL alternatives; Lemma Z also covers the stated bounded-dimension null-projection cases. EP with growing $p_{\mathcal{C}}$ and small residual without a proved size tail, approximate projection, non-Gaussian/heteroscedastic random omitted parts outside GR, random-design EP and n-dependent τ_n^2 are open; no density-contraction analog of Song–Liang Thm 4.
7. **Hellinger/predictive contraction and exponential rates** (Song–Liang Thms 1–2): not attempted (CJ20).
8. **Memory.** The memory envelope proved sufficient for delivery is $O((n+p)^2 + \mathrm{Cert}_M)$ words (C-DLV Lemma R; with MR1 (C-GROW), A1's M_G, same order); the shipped scorer is Gram-based (a Gram-free scorer is a declared, unshipped v3.2 item [SoP/02_method.tex:414-415]), so quadratic memory in p is a practical limit for $p \gg n$; no sub-quadratic bound is proved.
9. **Floating point.** Every theorem assumes EX-A (exact arithmetic, an idealisation). The only binary64 statement is Lemma UM-FP for the start-mode identification step, under FP-TAB, untested for the coded scorer; delivery also assumes IDEAL-W (the checked-overflow status `unsupported_size` fires for any fixed word size at large enough (n, p)).

10. **Implementation premises** remain premises: GUARD-W audit, integrated-LS0 replay, worker equivalence, IND-T extension ($R' > 4$), N9 START/KERNEL (candidate), e4 for the integrated LS0 centre (W1-N3: OPEN; used by the delivery results through TDP-IMPL, while the consistency results need only e4′ or 'some maximiser'). LS0, TDP-E, MR1 (C-GROW) and the fallback are implemented on the unmerged C-ENG branch (head `8a27652`), not in the frozen core or on main; at `8a27652` the branch preflight formula did not count the MR1 (C-GROW) LS0 term, while the current head `493579a` charges its worst case (see Remark (5) of §6). Song–Liang's Theorem S hypotheses were not inspected.

Recommended next steps (from the ticket reports). (i) Council scrutiny of Lemma 7.1 and a C-ENG confirmation of (L1)–(L3) on the merged code; (ii) a council pass for the corrected Proposition F (A1) and its GROW extension (A2/A3); (iii) GROW's frontier: NA with a low-rank common component and a polynomial ladder for factor designs; a sharper NA restricted to the path for finite-n onsets; (iv) C-MIS: the sufficient-statistic reduction for EP with small residual; (v) route (O-a)/(O-b) for the shipped calibration (pilot stage count, CJ06); (vi) HQ registry updates: N1 card status, TH161/TH163 rows, the patches of C-PACK.

9 Provenance

Council scores are (Codex gpt-6.1-sol / Claude Opus 5.5), round in which both seats passed with all gates. Review files are in each ticket's `reviews/`. Commits are on `main` (messages checked in the local reflog).

Result	proof files	review trail	council	commit
S-T / T03♯ (Thm 3.1)	SoP card TH09; `NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md`	SF W2-VT03R, XF A-REV2 (archive)	— (cited card)	frozen `21ebc28`
SL0, SL1, SL1′, SL2, map v2	`W1C/C-PACK/lemmas/I4_SL_corollaries.md`, `SONG_LIANG_MAP.md`	R34 (ERROR as written → 9 fixes), R38 CONTINUE, R39 sweep, C1	9.0 / 8.7 (r1)	`3b1d795`
e4′, Lemma UM, UM-FP	`W1C/C-PACK/lemmas/I1_e4_replacement.md`, `patches/P_e4_consumers.md`	R30 lock, R33 GAP → 7 fixes, R38, R39, C1	9.0 / 8.7 (r1)	`3b1d795`
LH-iid unconditional (Lemma I2)	`W1C/C-PACK/lemmas/I2_LHiid_unconditional.md`, `patches/P_GEN_LHiid.md`	R35 V-W-A (3 fixes), R38, R39, C1	9.0 / 8.7 (r1)	`3b1d795`
SEL, L8 (C1′) amendments	`W1C/C-PACK/patches/P_SEL_L6d.md`, `P_L8_certificate.md`	R31, R32 (xhigh) V-W-A; R37 re-check	9.0 / 8.7 (r1)	`3b1d795`
PXN consolidated supplement	`W1C/C-PACK/item5_PXN_consolidated/` (00–06, INDEX, MAP, APPLIED_ERRATA)	R36 fidelity FAITHFUL-WITH-FIXES; R39	(item 5 outside council scope)	`3b1d795`
Theorem PXN	`W1/W1-PXN/FINAL_PROOF.md`, `W1/W1-PXN/lemmas/`	R00 lock; R11–R17, R19, R21 item checks; R20 CONTINUE; R30 sweep (1 ERROR, 2 GAP, 4 MINOR fixed)	8.6 / 8.3 (r1)	`f1751f3`
W1-N3 (start mode as coded, TDP)	`W1/W1-N3/PROOF.md`, `W1/W1-N3/lemmas/`	R01 lock; R02, R03, R05 checks; R04 CONTINUE; R06 sweep	9.3 / 8.6 (r1)	`21d5e80`
Theorem GEN	`W1/W1-GEN/lemmas/00_MAIN.md`, L1–L8	R01 lock; R02–R04, R06–R09 checks; R05 CONTINUE; R10 sweep (0 fatal)	9.0 / 8.9 (r1)	`94ac305`
Lemma N1, Cor. E	`W1/W1-N1/lemmas/N1_theorem.md`, `E_corollary.md`, L1–L3	SL1 lock; LC1; MP1 CONTINUE; SW1 sweep (0 critical)	9.3 / 8.9 (r1)	`f16ce15`

Result	proof files	review trail	council	commit
Theorem MIS, F, Cor. H/D/D′, Lemma Z	`W1C/C-MIS/lemmas/00_MAIN.md`, L1–L6	R01 lock; R02, R03, R05 checks; R04 CONTINUE; R06 sweep (0 fatal)	9.2 / 9.0 (r1)	`05dcd85`
Theorem D, Cor. D′, (O), Prop. F (C-DLV form)	`W1C/C-DLV/lemmas/L1_resources.md–L5_insertions.md`, `INVENTORY.md`	R01 REJECT → v2; R02–R04 checks; R05 CONTINUE; R06 sweep; C1 not passed (7.6 / 8.6), fixed	9.0 / 8.9 (r2)	`604f58a`
Prop. F (corrected, Prop. 6.3)	`W1C/HQ_AUDIT/A1_propF_redteam.md`	HQ red team (Codex Sol xhigh): HOLDS-WITH-AMENDMENTS	round 3 PASSED (9.3 / 8.6), checked in full	`af2c454`
Theorem GROW	`W1C/C-GROW/lemmas/00_MAIN.md`, L0–L6	R01 lock; R02–R04, R06 (Opus), R07 checks; R05 CONTINUE; R08 sweep (0 fatal, 0 major)	9.2 / 9.0 (r1)	`1853fcd`
Prop. F on the GROW class (Prop. 6.4)	`W1C/HQ_AUDIT/A2_propF_GROW.md`	A2 Codex prover (EXTENDS); A3 cross-family check (Claude Opus) VERIFIED-WITH-AMENDMENTS, 5 MINOR applied	round 3 PASSED (9.3 / 8.6), checked in full	`a6097e7`
C-ENG implementation (LS0, TDP-E, MR1 (C-GROW) cap, tier-1 fallback)	branch `origin/w1c/ls0-merge`: `Theory_work/Wave1C/C-ENG/REPORT.md`, `tests/`, `numerics/`	code reviews R03, R04, R06 APPROVE; R05 APPROVE-WITH-FIXES (fixed); tests and 48/48 regression	— (engineering)	`8a27652`, `493579a` (reviewed / current head of the unmerged branch)
Lemma CAP (Lemma 7.1)	this chapter, §7.1	F1 fidelity reviewer (Codex Sol high) VERIFIED-WITH-AMENDMENTS, applied; council round 1: both seats found (a)–(c) correct (round not passed); A1 states the prefix property independently	round 3 PASSED (9.3 / 8.6), checked in full	`5bcde3b` (v1), `e95adc2` (v2), `d315c1e` (v3)

Sources: [W1C/C-PACK/REPORT.md:66-82]; [W1/W1-PXN/REPORT.md:71-82]; [W1/W1-GEN/REPORT.md:71-83]; [W1/W1-N1/REPORT.md:67-86]; [W1/W1-N3/REPORT.md:52-61]; [W1C/C-MIS/REPORT.md:116-129]; [W1C/C-DLV/REPORT.md:74-86]; [W1C/C-GROW/REPORT.md:72-82].